LIMIT

INTRODUCTION

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2021 - 2022

The tangent to a function

Consider the parabola

$$y = x^2$$

We are interested in the tangent to $f(x)$ at the point $P = (1, 1)$.

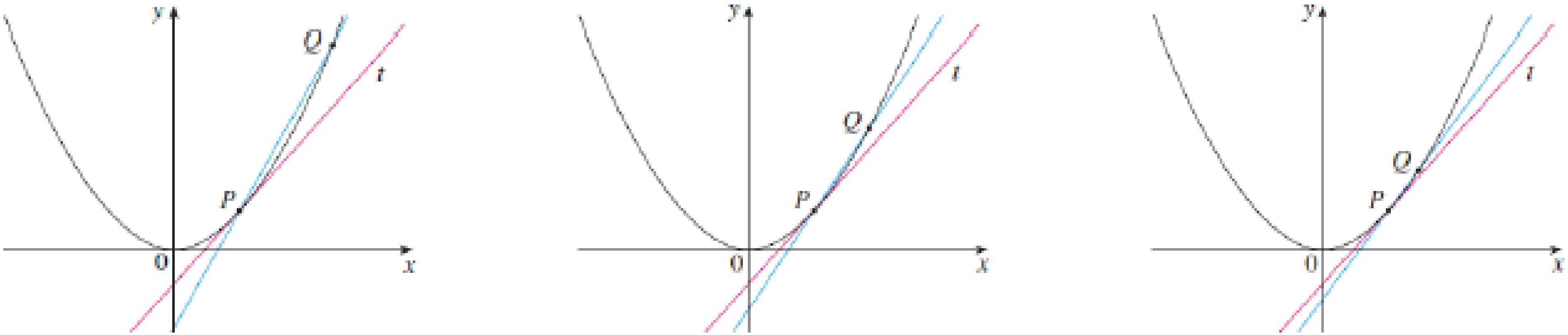
We take different values of Q (with $Q \neq 1$) and calculate the gradient of the straight line that goes through P and Q.

Values of m_{PQ}

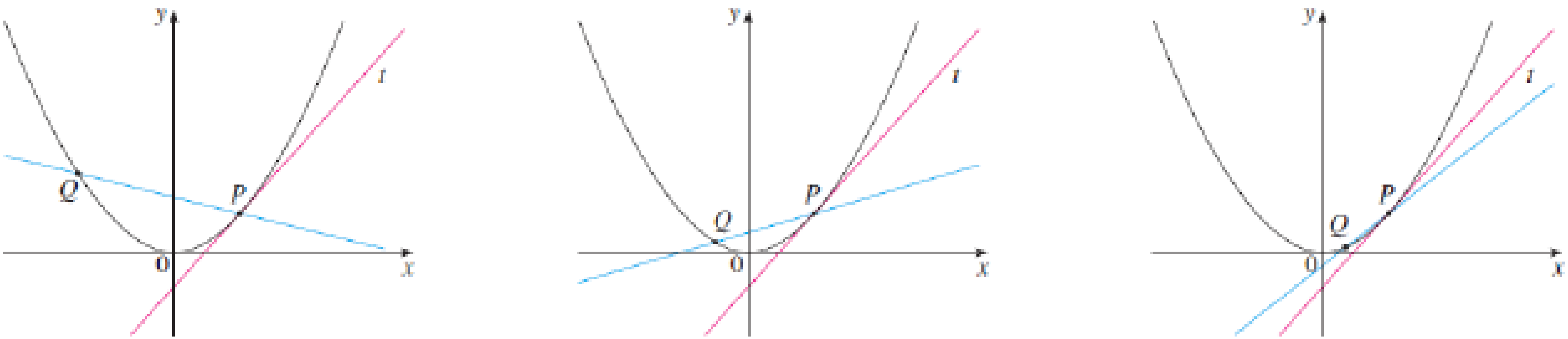
$x < 1$	m_{PQ}
0	1
0.5	1.5
0.9	1.9
0.99	1.99
0.999	1.999

$x > 1$	m_{PQ}
2	3
1.5	2.5
1.1	2.1
1.01	2.01
1.001	2.001

The tangent line as Q approaches P



Q approaches P from the right

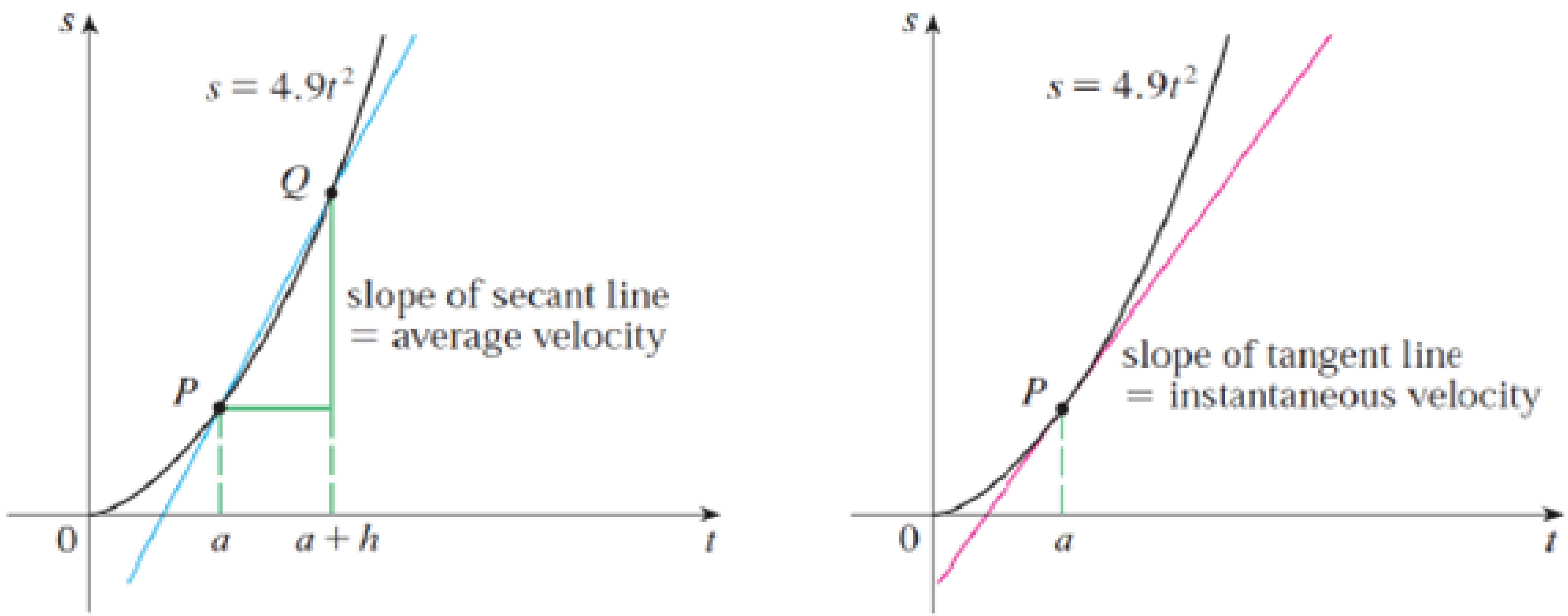


Q approaches P from the left

Velocity

We can easily measure distance with respect to time, but to measure instantaneous velocity we need to use limits.

Dropping a ball from a tower



From tangents to arbitrary functions

In the tangent problem, when we take $x = 1$ we cannot calculate the slope of the tangent because

$$m_{PQ} = \frac{y_P - y_Q}{x_P - x_Q} = \frac{1-1}{1-1} \quad \text{does not exist}$$

The tangent and velocity problems are examples of when we might want to calculate the *limit* of the value of a function. We now consider the definition of a limit for an arbitrary function f.

LIMIT

DEFINITION

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Definition of a limit of a function

Definition: Suppose $f(x)$ is defined for x near* a. If we can get the values of $f(x)$ as close to L as we like by taking x as close to a on either side (but not equal to a), then we write

$$\lim_{x \to a} f(x) = L$$

and say

"the limit of $f(x)$, as x approaches a, equals L"

*near: $f(x)$ is defined on some open interval including a, but possibly not at a itself.

Example

Estimate the value of $\lim_{x \to 1} \dfrac{x-1}{x^2-1}$.

$x < 1$	$f(x)$
0.5	0.666667
0.9	0.526316
0.99	0.502513
0.999	0.500250
0.9999	0.500025

$x > 1$	$f(x)$
1.5	0.4
1.1	0.476190
1.01	0.497512
1.001	0.499750
1.0001	0.499975

Based on the above calculations, we estimate that

$$\lim_{x \to 1} \frac{x-1}{x^2-1} = \frac{1}{2}.$$

$$g(x) = \begin{cases} \dfrac{x-1}{x^2-1} & \text{if } x \neq 1 \\ 2 & \text{if } x = 1 \end{cases}$$

This new function g still has the same limit as x approaches 1. (See Figure 4.)

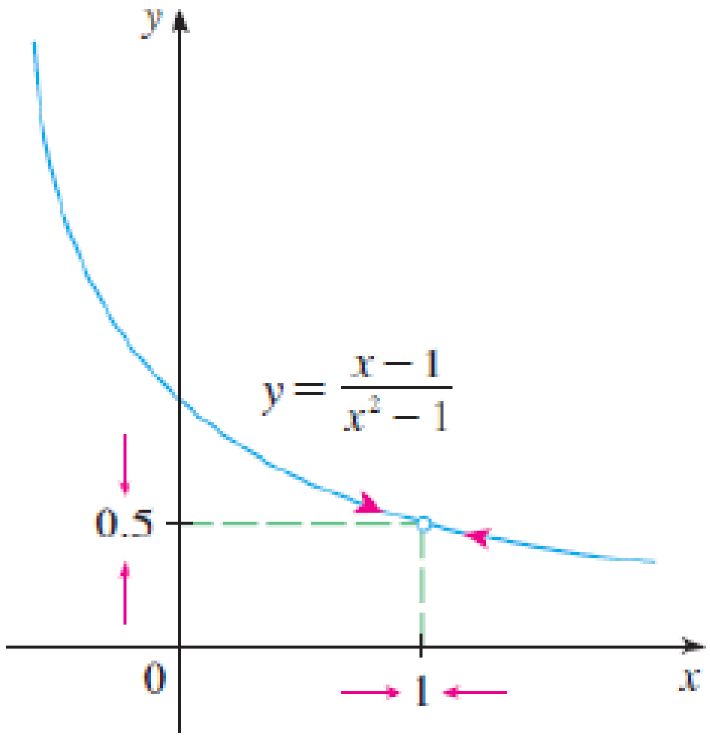


FIGURE 3

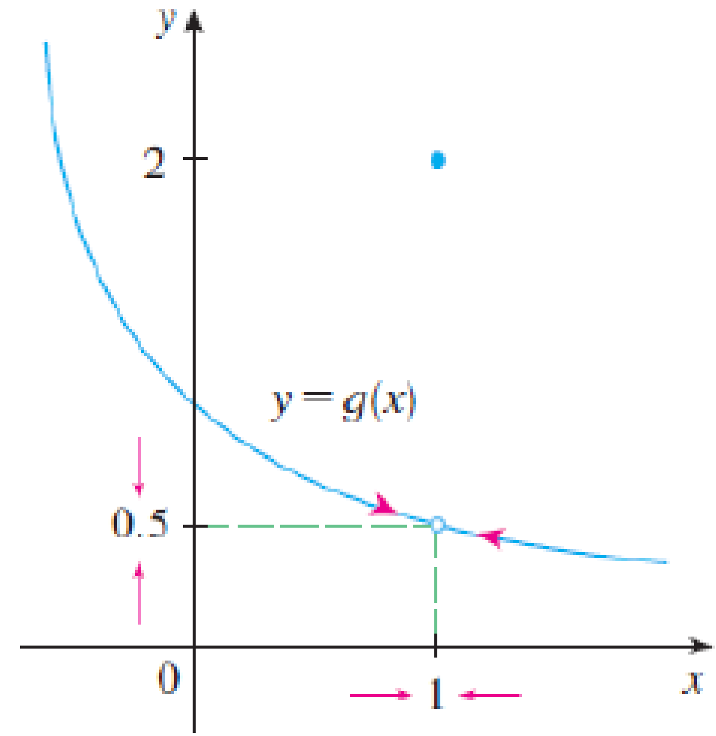


FIGURE 4

Example:

Consider the function $f(x) = \dfrac{x^2 + 4x + 4}{x + 2}$.

Observe that

$$f(x) = \frac{(x+2)(x+2)}{x+2}$$

Is this different from the function $g(x) = x + 2$?

Example

$$g(x) = \begin{cases} x+3 & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

Sketch the function and find $\lim\limits_{x\to 2} g(x)$.

One-sided limits

Definition: We write

$$\lim_{x \to a^-} f(x) = L$$

and say "the limit of $f(x)$ as x approaches a from the left" is equal to L if we can make the values of $f(x)$ as close as we like to L by taking x close to a, but with x less than a.

This is also called "the left-hand limit of $f(x)$ as x approaches a"

From the other side

Definition: We write

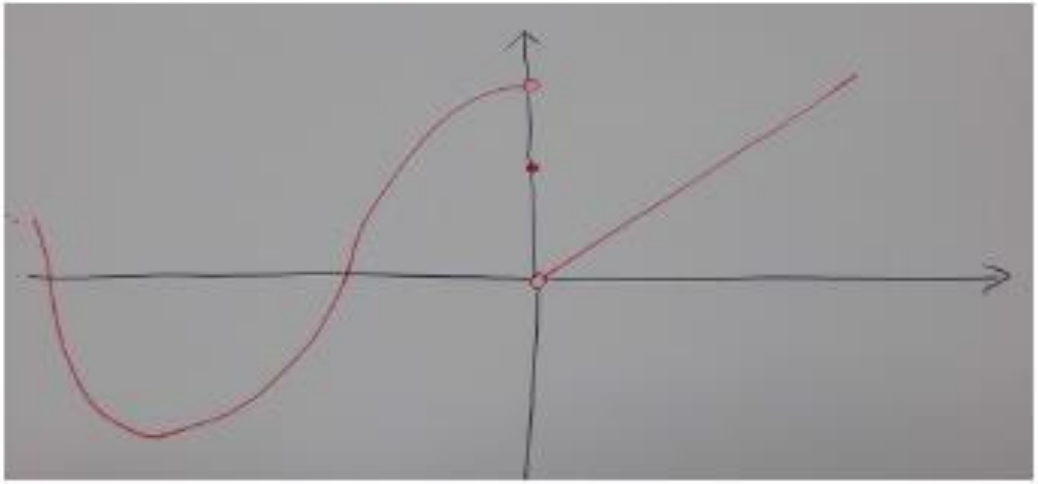
$$\lim_{x \to a^+} f(x) = L$$

and say "the limit of $f(x)$ as x approaches a from the right" is equal to L if we can make the values of $f(x)$ as close as we like to L by taking x close to a, but with x greater than a.

This is also called "the right-hand limit of $f(x)$ as x approaches a"

One-sided example

$$h(x) = \begin{cases} \cos x & \text{if } x < 0 \\ \frac{1}{2} & \text{if } x = 0 \\ x & \text{if } x > 0 \end{cases}$$



$$\lim_{x \to 0^-} h(x) = 1, \; \lim_{x \to 0^+} h(x) = 0, \; h(0) = \frac{1}{2}$$

One (sided) + one (sided) = two (sided)

For $L \in \mathbb{R}$ we have the following:

$$\lim_{x \to a} f(x) = L \quad \text{if and only if}$$

$$\lim_{x \to a^-} f(x) = L \quad \text{and} \quad \lim_{x \to a^+} f(x) = L$$

If there exists $L \in \mathbb{R}$ such that $\lim_{x \to a} f(x) = L$ then we say:

"the limit of $f(x)$ as x approaches a exists"

One (sided) + one (sided) = two (sided)

In the previous two examples of piece-wise functions we have:

- $\lim_{x \to 2} g(x)$ exists

  (since $\lim_{x \to 2^-} g(x) = 5 = \lim_{x \to 2^+} g(x)$)

- $\lim_{x \to 0} h(x)$ does not exist

  (because $\lim_{x \to 0^-} h(x) \neq \lim_{x \to 0^+} h(x)$)

LIMIT

FORMAL DEFINITION

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Definition of a limit of a function

Definition: Suppose $f(x)$ is defined for x near* a. If we can get the values of $f(x)$ as close to L as we like by taking x as close to a on either side (but not equal to a), then we write

$$\lim_{x \to a} f(x) = L$$

and say

"the limit of $f(x)$, as x approaches a, equals L"

*near: $f(x)$ is defined on some open interval including a, but possibly not at a itself.

Before, we give the formal definition, let us think of the previous definition in a more informal way.

Essentially, the definition of a limit is saying:

If you give me any y value close to L, I can give you an x value such that $f(x)$ will be closer than your y value to L.

The importance of absolute values

In our first definition we said

"as close to L as we like"

and on the last slide we said

"a y value close to L"

Q: Do we want to the y value to be a little bit below L or a little bit above L?

A: Both!

It shouldn't matter if the y value is bigger or smaller than L. What matters is the **distance from y to L.**

Two important characters

ε "epsilon" and δ "delta"

These two Greek letters are often used in mathematics to represent any small positive real number.

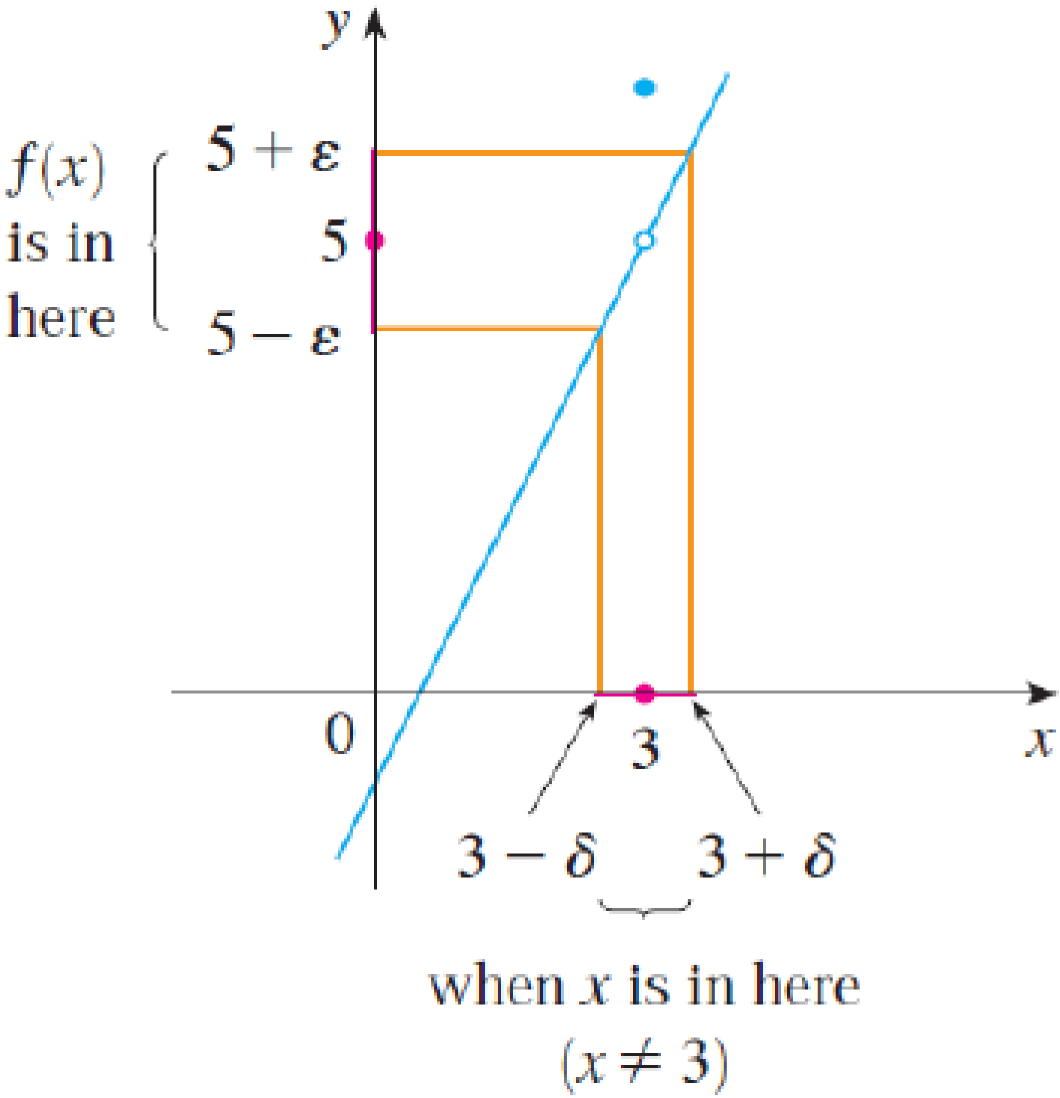
By small, we usually mean "close to 0".

Definition:

We write $\lim_{x \to a} f(x) = L$ if the following holds:

for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $0 < |x - a| < \delta$ then $|f(x) - L| < \varepsilon$.

This is saying for any $\varepsilon > 0$, there exists a $\delta > 0$ (which depends on ε) such that whenever x is within δ from a, then $f(x)$ will be within ε from L.



$$\lim_{x \to 3} f(x) = 5$$

Let

$$f(x) = \begin{cases} \frac{x}{3} + 6 & \text{if } x \neq 3 \\ 6 & \text{if } x = 3 \end{cases}$$

and let us now prove that

$$\lim_{x \to 3} f(x) = 7$$

Understanding the formal definition

Tools to enrich calculus

Click on "Limits and Derivatives" and then "Precise Definitions of Limits". Move ε to a value greater than zero and then see what δ value gives you an interval $(a-\delta, a+\delta)$ for which the distance from $f(x)$ to L will always be less than ε (i.e. $|f(x)-L|<\varepsilon$).

Note that

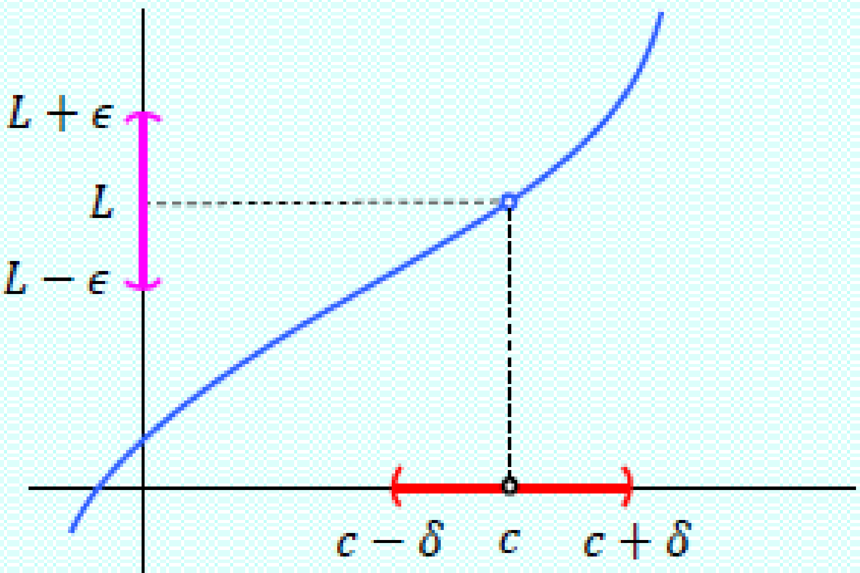
$$x \in (a-\delta, a+\delta) \iff |x-a|<\delta.$$

Pembahasan limit secara formal matematika merupakan salah satu topik yang **sangat sangat sangat sulit** untuk dicerna. Namun demikian, konsep ini merupakan bagian yang **sangat penting**. Mengapa penting? Karena dengan konsep ini kita dapat menjustifikasi kebenaran logika pada rumus-rumus dan perhitungan limit.

Sebelum memasuki pembahasan limit secara formal, anda diharapkan dapat memahami/merasakan/meresapi arti dari notasi berikut,

$$0 < |x - c| < \delta \quad \text{dan} \quad |f(x) - L| < \epsilon$$

Kedua hal ini diilustrasikan melalui gambar berikut ini:



$$0 < |x - c| < \delta$$

$$\Updownarrow$$

$$c - \delta < x < c + \delta, \quad x \neq c$$

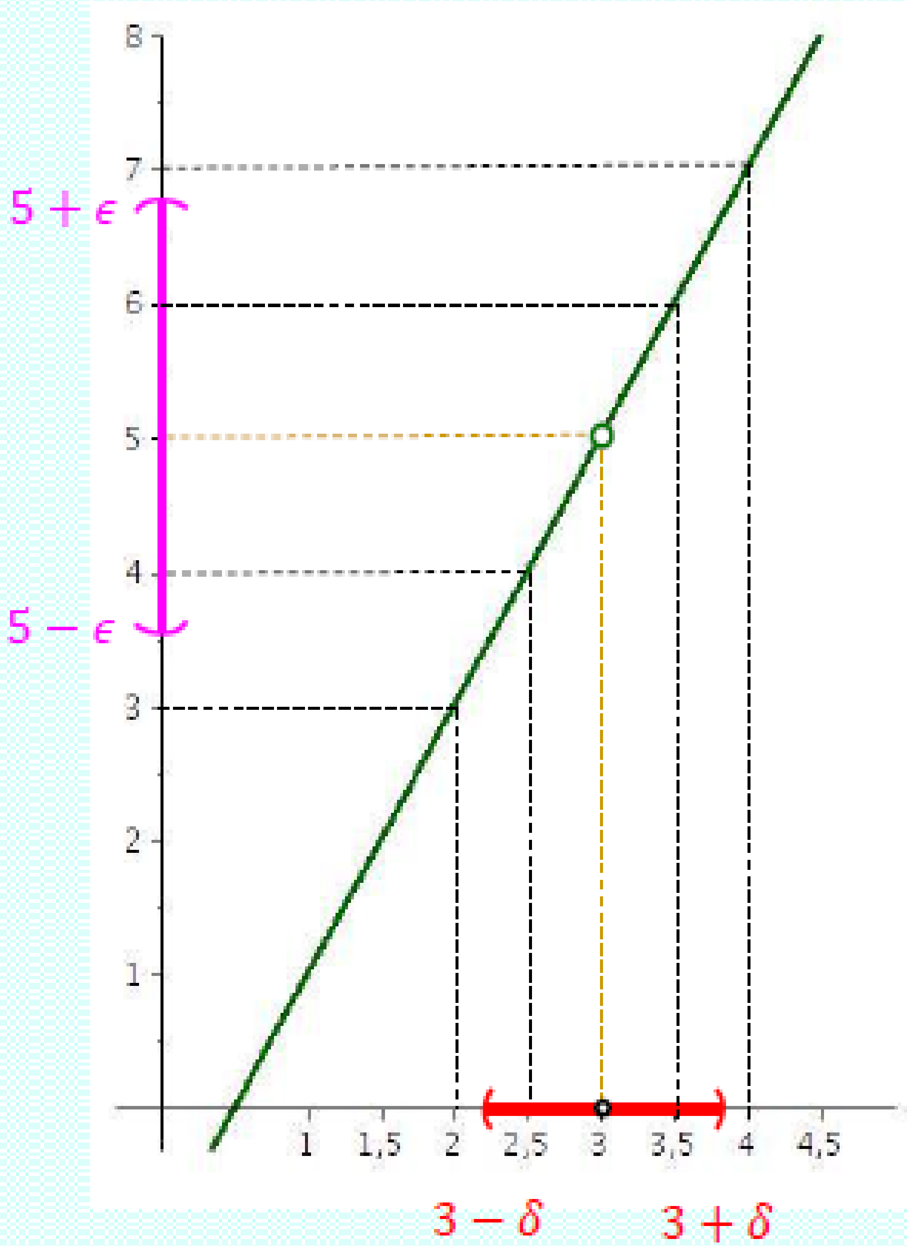
Berikut ini disajikan contoh kalkulasi $\epsilon - \delta$ yang akan mengantarkan kita pada pendefinisian konsep limit secara formal.

Diberikan fungsi $f(x) = 2x - 1,\ x \neq 3$

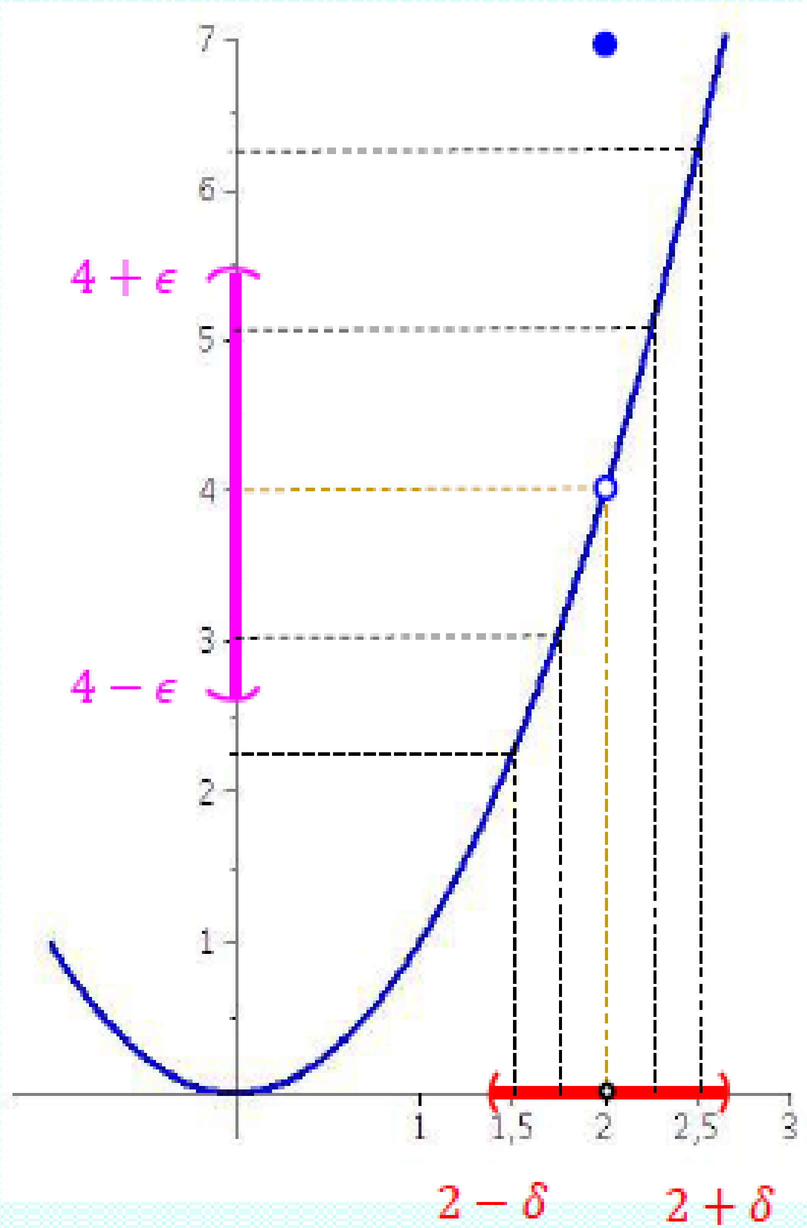
Untuk setiap nilai ϵ pada table di bawah ini, tentukanlah $\delta > 0$, supaya

$0 < |x - 3| < \delta \implies |f(x) - 5| < \epsilon$

ϵ	δ
1	Δ
0,1	
0,01	
0,001	
0,0001	



Berikut ini disajikan contoh kalkulasi $\epsilon - \delta$ yang akan mengantarkan kita pada pendefinisian konsep limit secara formal.



Diberikan fungsi $f(x) = \begin{cases} x^2, & x \neq 2 \\ 7, & x = 2 \end{cases}$

Untuk setiap nilai ϵ pada table di bawah ini, tentukanlah $\delta > 0$, supaya

$$0 < |x-2| < \delta \Rightarrow |f(x) - 4| < \epsilon$$

ϵ	δ
1	Δ
0,1	
0,01	
0,001	
0,0001	

Buktikan $\lim_{x \to 2} 3x + 2 = 8$.

- Tetapkan $\epsilon > 0$, akan dicari $\delta > 0$ sehingga $0 < |x - 2| < \delta \Rightarrow |f(x) - 8| < \epsilon$
- $|f(x) - 8| < \epsilon \Leftrightarrow |3x + 2 - 8| < \epsilon$

$$\Leftrightarrow |3x - 6| < \epsilon$$

$$\Leftrightarrow -\epsilon < 3x - 6 < \epsilon$$

$$\Leftrightarrow 6 - \epsilon < 3x < 6 + \epsilon$$

$$\Leftrightarrow 2 - \frac{\epsilon}{3} < x < 2 + \frac{\epsilon}{3}$$

- $|f(x) - 8| < \epsilon \Leftrightarrow -\frac{\epsilon}{3} < x - 2 < \frac{\epsilon}{3}$

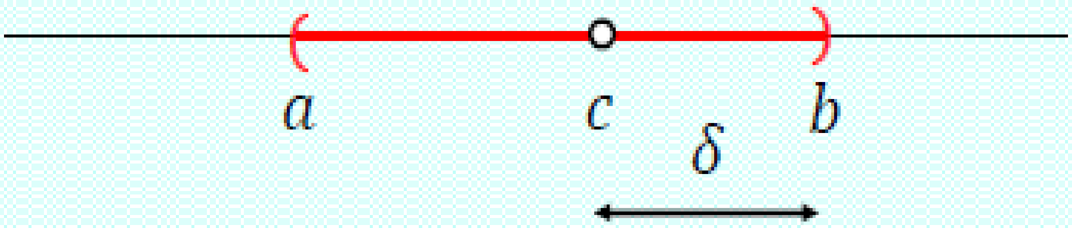
$$\Leftrightarrow |x - 2| < \frac{\epsilon}{3}$$

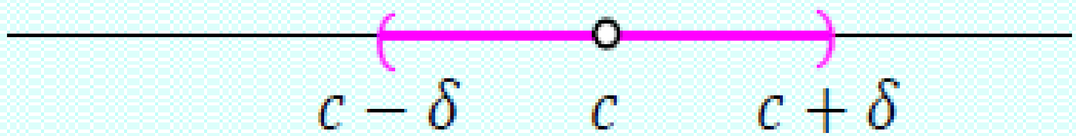
Jadi ambil $\delta = \frac{\epsilon}{3}$, maka $0 < |x - 2| < \delta \Rightarrow |f(x) - 8| < \epsilon$ dipenuhi.

Dalam pembuktian $\lim_{x \to c} f(x) = L$, kita harus mencari $\delta > 0$ supaya memenuhi $0 < |x - c| < \delta \Rightarrow |f(x) - L| < \epsilon$.

Berikut ini disajikan langkah-langkah umum yang dapat kita lakukan:

1. Selesaikan pertaksamaan $|f(x) - L| < \epsilon$ untuk memeperoleh interval (a, b) yang memuat titik c, dimana semua titik x pada $(a, b) \backslash \{c\}$ memenuhi pertaksamaan tersebut.





2. Tetapkan $\delta = \min\{c - a, b - c\}$, maka setiap titik $x \in (c - \delta, c + \delta) \backslash \{c\}$ akan memenuhi $|f(x) - L| < \epsilon$ (mengapa?)

Buktikan $\lim_{x \to 2} x^2 = 4$.

- Tetapkan $\epsilon > 0$, akan dicari $\delta > 0$ sehingga $0 < |x-2| < \delta \Rightarrow |f(x) - 4| < \epsilon$
- $|f(x) - 4| < \epsilon \Leftrightarrow |x^2 - 4| < \epsilon \Leftrightarrow -\epsilon < x^2 - 4 < \epsilon \Leftrightarrow 4 - \epsilon < x^2 < 4 + \epsilon \quad (^*)$

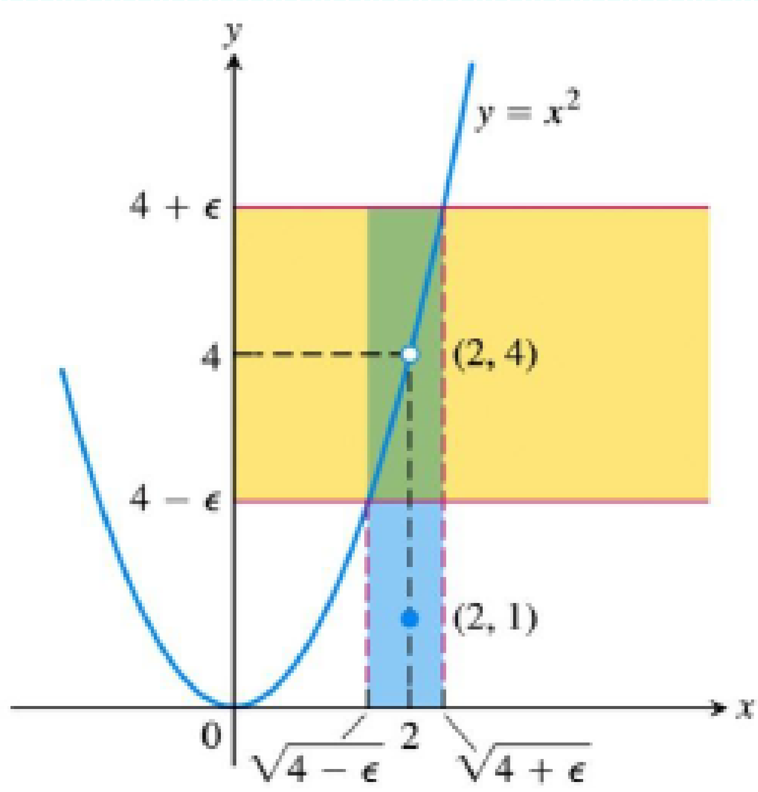
Karena $4 - \epsilon$ bisa negatif, kita bagi kasusnya menjadi dua bagian

Untuk $\epsilon < 4$, dari (*)

$$|f(x) - 4| \Leftarrow \sqrt{4-\epsilon} < x < \sqrt{4+\epsilon}$$

Kita harus menentukan $\delta > 0$, supaya

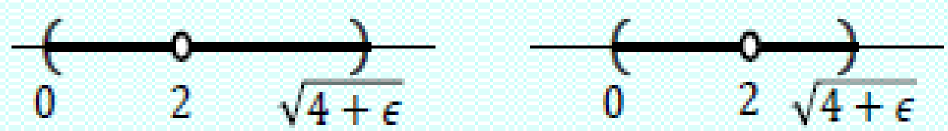
$$(2-\delta, 2+\delta) \subset (\sqrt{4-\epsilon}, \sqrt{4+\epsilon})$$



Ambil $\delta = \min\{2 - \sqrt{4-\epsilon}, \sqrt{4+\epsilon} - 2\}$

Untuk $\epsilon \geq 4$, dari (*)

$$|f(x) - 4| \Leftarrow 0 \leq x < \sqrt{4+\epsilon}$$ mengapa?



Untuk itu pilih $\delta = \min\{2, \sqrt{4+\epsilon} - 2\}$

LIMIT

LIMIT LAWS

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Limit Laws

Suppose that c is a constant and the limits $\lim_{x \to a} f(x)$ and $\lim_{x \to a} g(x)$ exist. Then

1. $\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$
2. $\lim_{x \to a} [f(x) - g(x)] = \lim_{x \to a} f(x) - \lim_{x \to a} g(x)$
3. $\lim_{x \to a} [cf(x)] = c \lim_{x \to a} f(x)$
4. $\lim_{x \to a} [f(x)g(x)] = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$
5. $\lim_{x \to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}$ if $\lim_{x \to a} g(x) \neq 0$

6. $\lim_{x \to a} [f(x)]^n = \left[\lim_{x \to a} f(x)\right]^n$

7. $\lim_{x \to a} c = c$

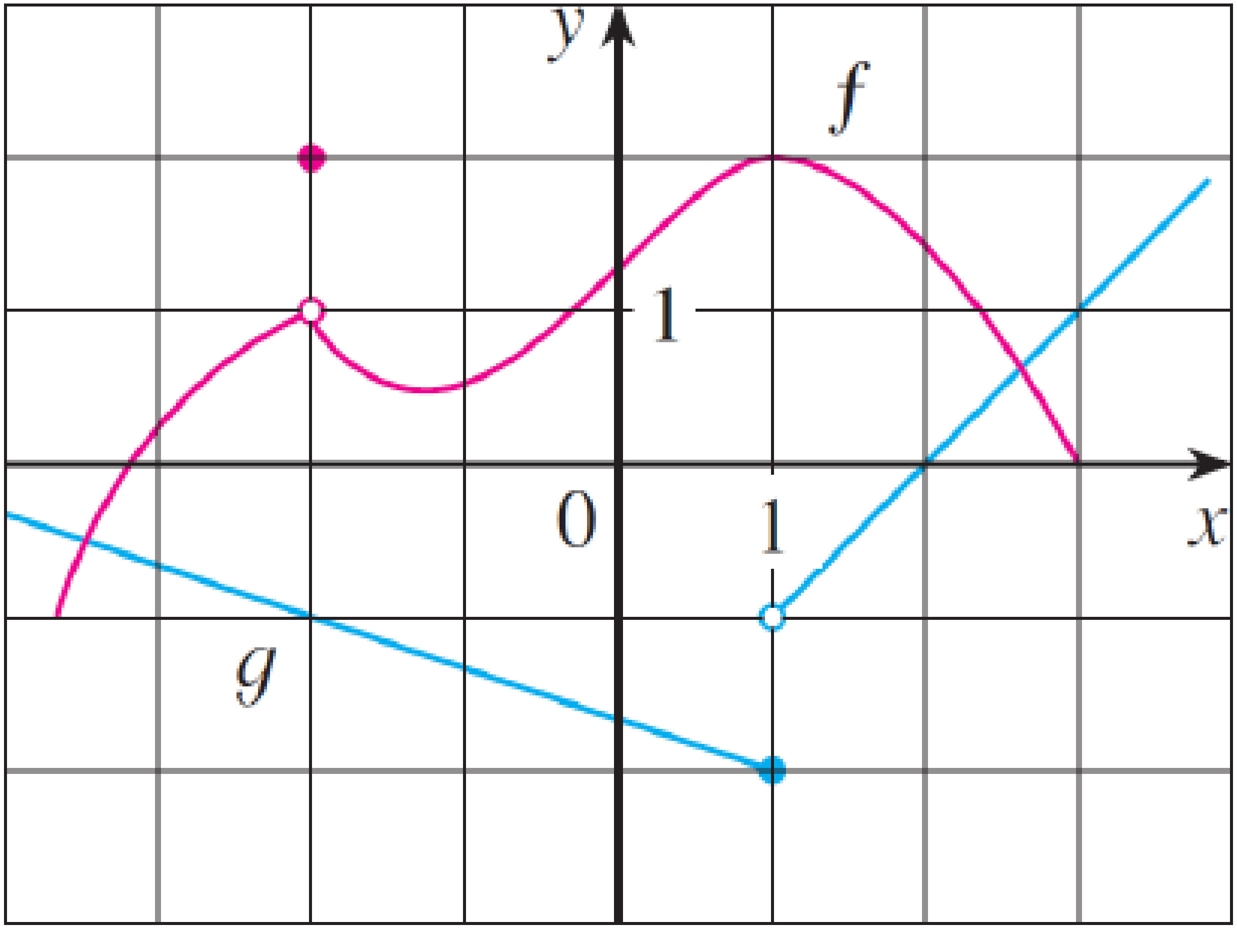
8. $\lim_{x \to a} x = a$

9. $\lim_{x \to a} x^n = a^n$ where $n \in \mathbb{Z}^+$

10. $\lim_{x \to a} \sqrt[n]{x} = \sqrt[n]{a}$ where $n \in \mathbb{Z}^+$

11. $\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)}$ $(n \in \mathbb{Z}^+)$

Note: if n is even we must assume that $\lim_{x \to a} f(x) > 0$.



Evaluate the following limits if they exist:

$$\lim_{x \to -2} [f(x) + g(x)] \quad \lim_{x \to 1} [f(x)g(x)] \quad \lim_{x \to 2} \frac{f(x)}{g(x)}$$

Example

Calculate the following limit using the limit laws:

$$\lim_{x \to -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$$

Substitution

<u>Direct Substitution Property:</u>

If f is a polynomial or a rational function and a is in the domain of f, then

$$\lim_{x \to a} f(x) = f(a)$$

How do we get this? You can prove it using Limit Laws 1–9. How would you do this? By induction on the degree of the polynomial(s).

Example

Calculate the following limit using the limit laws:

$$\lim_{x \to -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$$

Useful fact:

If $f(x) = g(x)$ when $x \neq a$, then

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x),$$

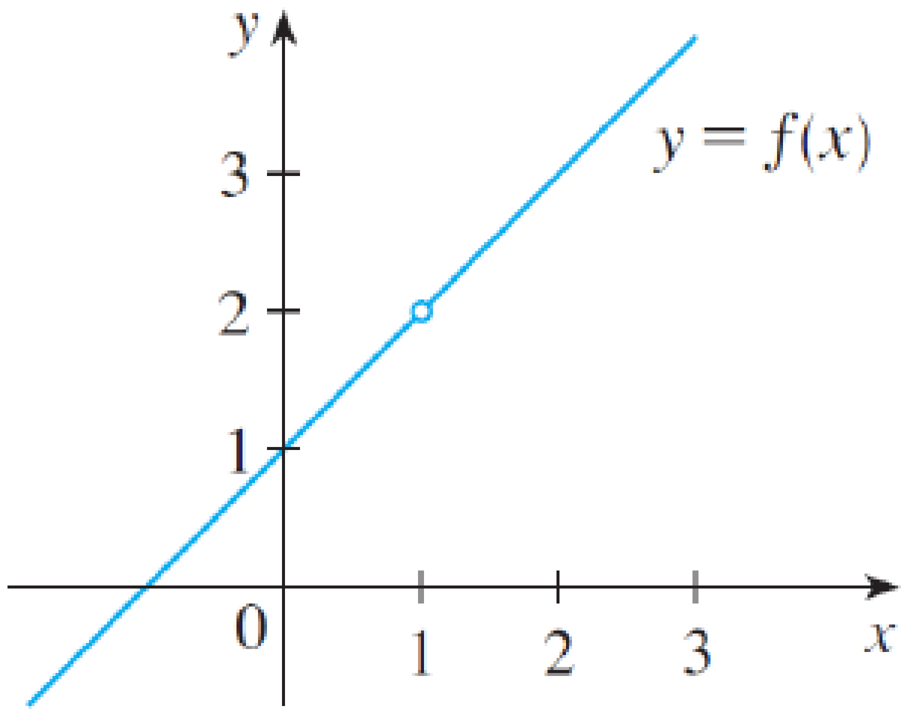
provided that both limits exist.

We will use this fact to 'cancel' factors so that we can then make direct substitutions.

Example: $\dfrac{(x+1)(x-1)}{x-1} = x+1$ when $x \neq 1$.

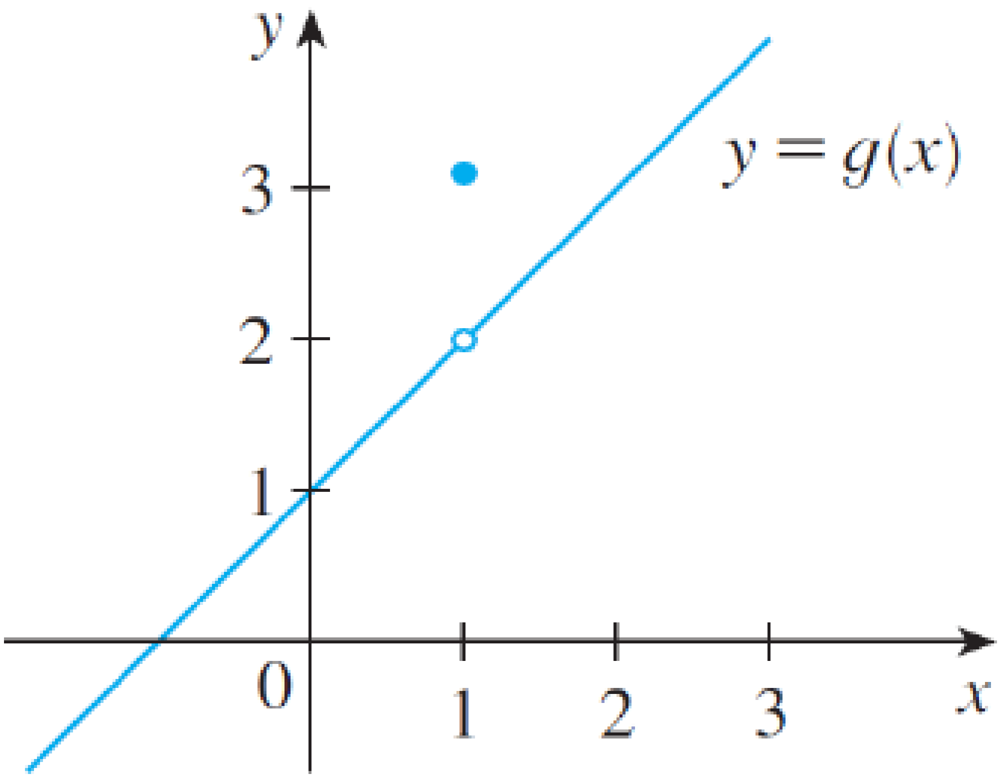
Now we use the fact from the previous slide:

$$\lim_{x\to 1} \frac{x^2-1}{x-1} = \lim_{x\to 1} \frac{(x+1)(x-1)}{x-1} = \lim_{x\to 1}(x+1) = 2$$



Example

$$\lim_{x\to 1} g(x) \text{ where } g(x) = \begin{cases} x+1 & \text{if } x \neq 1 \\ \pi & \text{if } x = 1 \end{cases}$$



Examples:

Evaluate:

$$\lim_{h\to 0} \frac{(3+h)^2 - 9}{h}$$

$$\lim_{t\to 0} \frac{\sqrt{t^2+9} - 3}{t^2}$$

In each of the examples above we cannot use direct substitution as the functions are undefined at $h = 0$ and $t = 0$. We must change the functions into new functions that agree everywhere except at 0.

More examples:

- $$\lim_{x\to 16} \frac{4-\sqrt{x}}{16x-x^2}$$
- $$\lim_{t\to 0} \left(\frac{1}{t} - \frac{1}{t^2+t}\right)$$

Calculating one-sided limits

First, a reminder about the existence of a two-sided limit:

Theorem: $\lim_{x \to a} f(x) = L$ if and only if

$$\lim_{x \to a^-} f(x) = L = \lim_{x \to a^+} f(x)$$

Example: Show that $\lim_{x \to 0} \frac{|x|}{x}$ does not exist.

Example: Show that $\lim_{x\to 0} \frac{|x|}{x}$ does not exist.

To do this we must calculate both

$$\lim_{x\to 0^-} \frac{|x|}{x} \qquad \text{and} \qquad \lim_{x\to 0^+} \frac{|x|}{x}.$$

Example:

Let

$$f(x) = \begin{cases} \sqrt{x-4} & \text{if } x > 4 \\ 8 - 2x & \text{if } x < 4 \end{cases}$$

Determine if $\lim_{x \to 4} f(x)$ exists.

Inequalities and Limits

Theorem: If $f(x) \leqslant g(x)$ when x is near a (except possibly at a) and the limits of f and g both exist as x approaches a, then

$$\lim_{x \to a} f(x) \leqslant \lim_{x \to a} g(x)$$

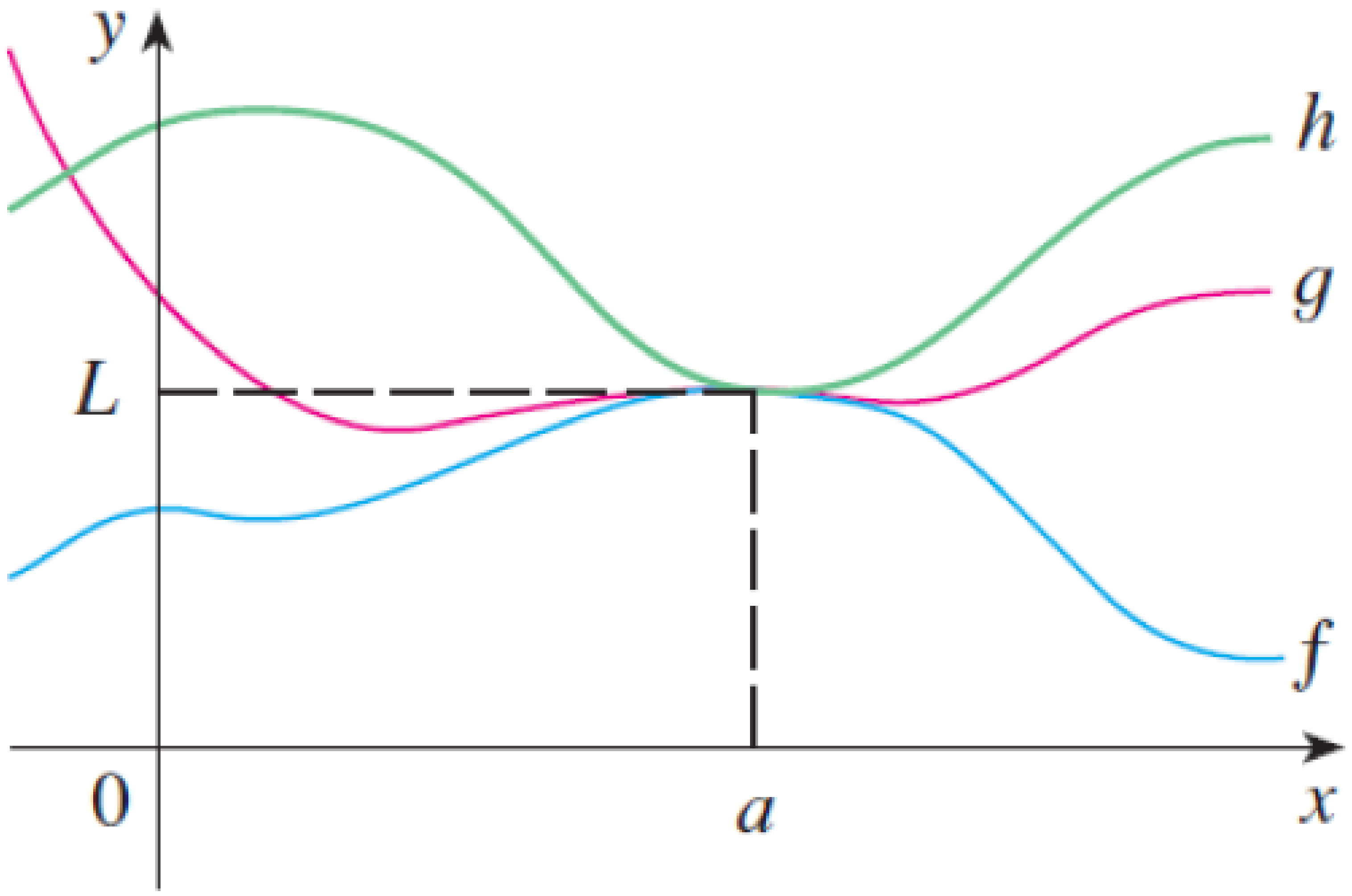
The Squeeze Theorem

If $f(x) \leqslant g(x) \leqslant h(x)$ when x is near a (except possibly at a) and

$$\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L,$$

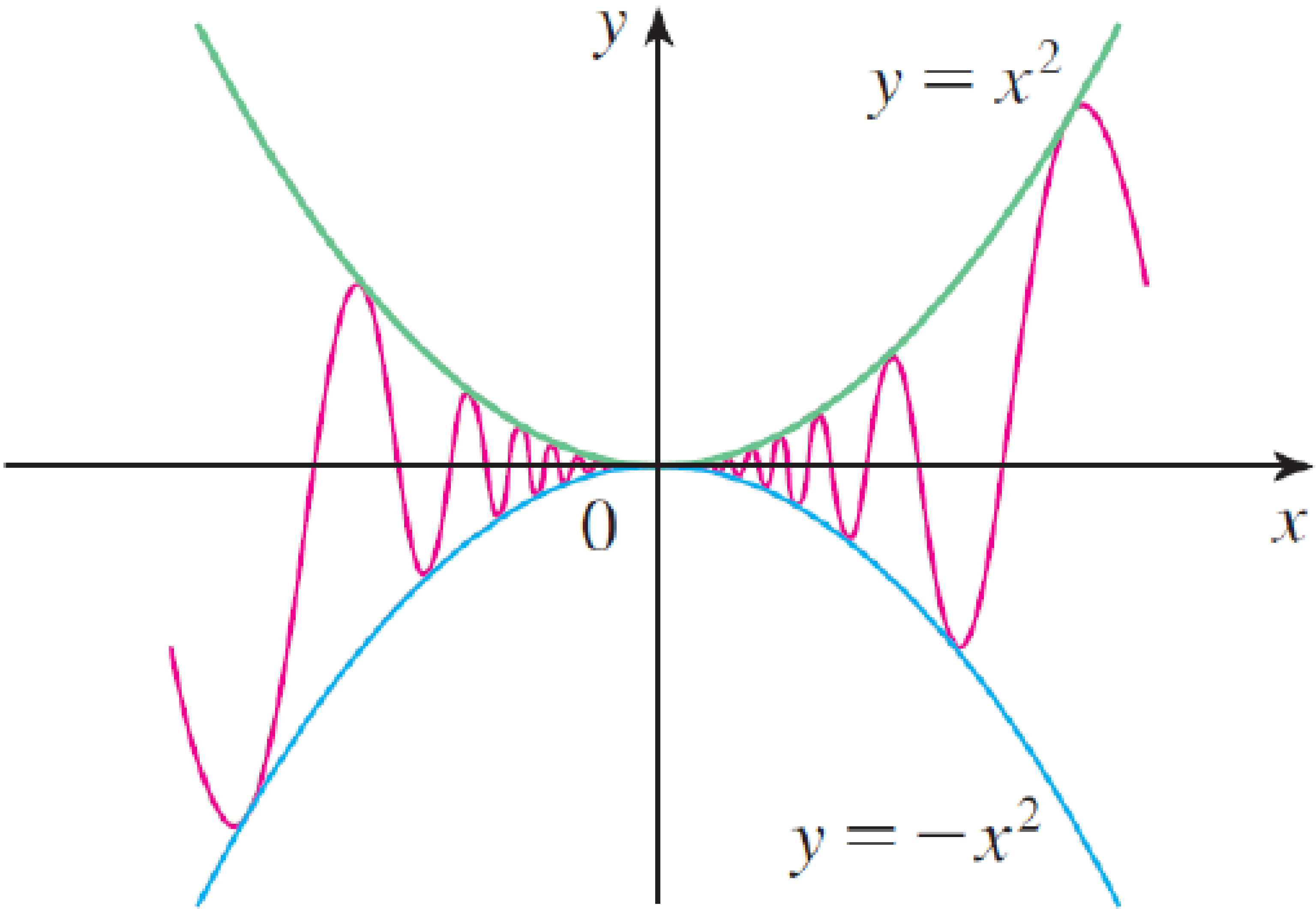
then

$$\lim_{x \to a} g(x) = L.$$

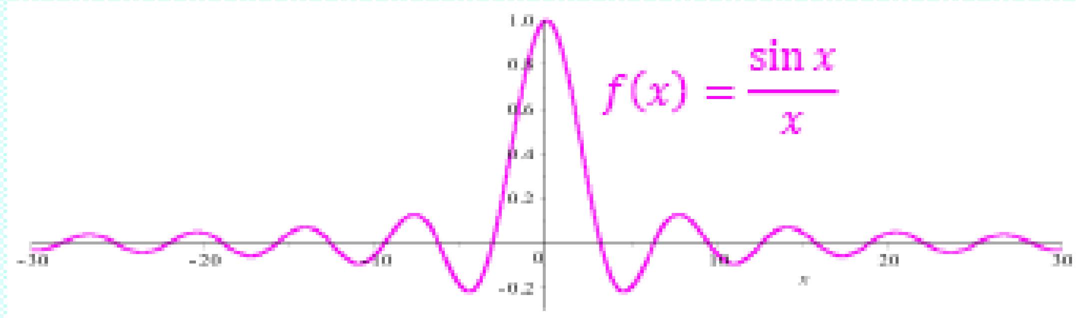


y
h
g
L
f
0
a
x

Example: show that $\lim_{x \to 0} x^2 \sin\left(\frac{1}{x}\right) = 0$.



Teorema Limit Fungsi Trigonometri



a. $\lim\limits_{x \to c} \sin x = \sin c$

b. $\lim\limits_{x \to c} \cos x = \cos c$

c. $\lim\limits_{x \to 0} \frac{\sin x}{x} = 1$ Δ, tetapi hati-hati, bila $c \neq 0$, $\lim\limits_{x \to c} \frac{\sin x}{x}$ belum tentu 1

d. $\lim\limits_{x \to 0} \frac{x}{\sin x} = 1$

e. $\lim\limits_{x \to 0} \frac{\tan x}{x} = 1$

f. $\lim\limits_{x \to 0} \frac{x}{\tan x} = 1$

- $\lim\limits_{x \to 1} \frac{\sin(x-1)}{x-1} = 1$
- $\lim\limits_{x \to 1} \frac{\sin(x-1)^2}{(x-1)^2} = 1$
- $\lim\limits_{x \to 0} \frac{\sin(x-1)}{x-1}$ bukan 1 melainkan $\frac{\sin(-1)}{-1}$
- $\lim\limits_{x \to 0} \frac{\sin\frac{1}{x}}{\frac{1}{x}}$ bukan 1

Hitung limit-limit berikut dan jelaskan keabsahan pada tiap langkahnya.

1. $$\lim_{x\to 3}\frac{x^4-3x}{3x^2-5x+7} \quad \underline{\Psi}$$

2. $$\lim_{x\to 3}\frac{x^2-2x-3}{x-3} \quad \underline{\Psi}$$

3. $$\lim_{x\to 1}\frac{2x^3+3x^2-2x-3}{x^2-1} \quad \underline{\Psi}$$

4. $$\lim_{x\to 0}\frac{x-\sin(2x)}{2x+\tan x} \quad \underline{\Psi}$$

5. $$\lim_{x\to \frac{1}{2}\pi}\left(x-\frac{1}{2}\pi\right)\tan(3x) \quad \underline{\Psi}$$

6. $$\lim_{x\to \pi}\frac{1+\cos x}{\sin(2x)} \quad \underline{\Psi}$$

7. $$\lim_{x\to 0} x\sin\left(\frac{1}{x}\right) \quad \underline{\Psi}$$

8. $$\lim_{x\to 0} x^2\cos\left(\frac{1}{x}\right) \quad \underline{\Psi}$$

LIMIT

INFINITE LIMIT

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Infinite limits:

Definition: Let f be a function defined on both sides of a, except possibly at a itself. Then

$$\lim_{x \to a} f(x) = \infty$$

means that $f(x)$ can get as big as we like by taking x sufficiently close to a.

Example: $\displaystyle\lim_{x \to 0} \frac{1}{x^2} = \infty$

Infinite Limits

Previously we said that $f(x)$ has an infinite limit as $x \to a$ if we can make $f(x)$ "as big as we like" by taking x close to a.

Now we want to formalise this in the same way that we formalised the definition of a (non-infinite) limit.

Definition: Let f be a function defined on some open interval that contains a, except possibly at a itself. Then

$$\lim_{x \to a} f(x) = \infty$$

means that for every positive number M there is a $\delta > 0$ such that

if $0 < |x - a| < \delta$ then $f(x) > M$.

<u>Definition:</u> Let f be a function defined on some open interval that contains a, except possibly at a itself. Then

$$\lim_{x \to a} f(x) = -\infty$$

means that for every negative number N there is a $\delta > 0$ such that

if $0 < |x - a| < \delta$ then $f(x) < N$.

One-sided infinite limits:

Let $f(x) = \dfrac{1}{x}$,
and consider

$$\lim_{x \to 0^-} f(x) \qquad \text{and} \qquad \lim_{x \to 0^+} f(x)$$

Let $g(x) = \csc x$, and consider

$$\lim_{x \to \pi^-} g(x) \qquad \text{and} \qquad \lim_{x \to \pi^+} g(x)$$

Very important:

For the limit at x approaches a of $f(x)$ to exist, we need

$$\lim_{x \to a} f(x) = L$$

for some $L \in \mathbb{R}$.

Therefore, $\displaystyle\lim_{x \to 0} \frac{1}{x^2}$ does not exist.

Example: find the following limits:

$$\lim_{x\to 3^+} \frac{2x}{x-3} \qquad \text{and} \qquad \lim_{x\to 3^-} \frac{2x}{x-3}$$

To calculate the first limit, consider an x value just a little bit bigger than 3 (e.g. 3.1) and then see what happens as you get closer to 3 (e.g. 3.01).
To calculate the second limit, consider an x value slightly smaller than 3 (e.g. 2.9) and then get closer to 3 (e.g. 2.99).

Vertical asymptotes

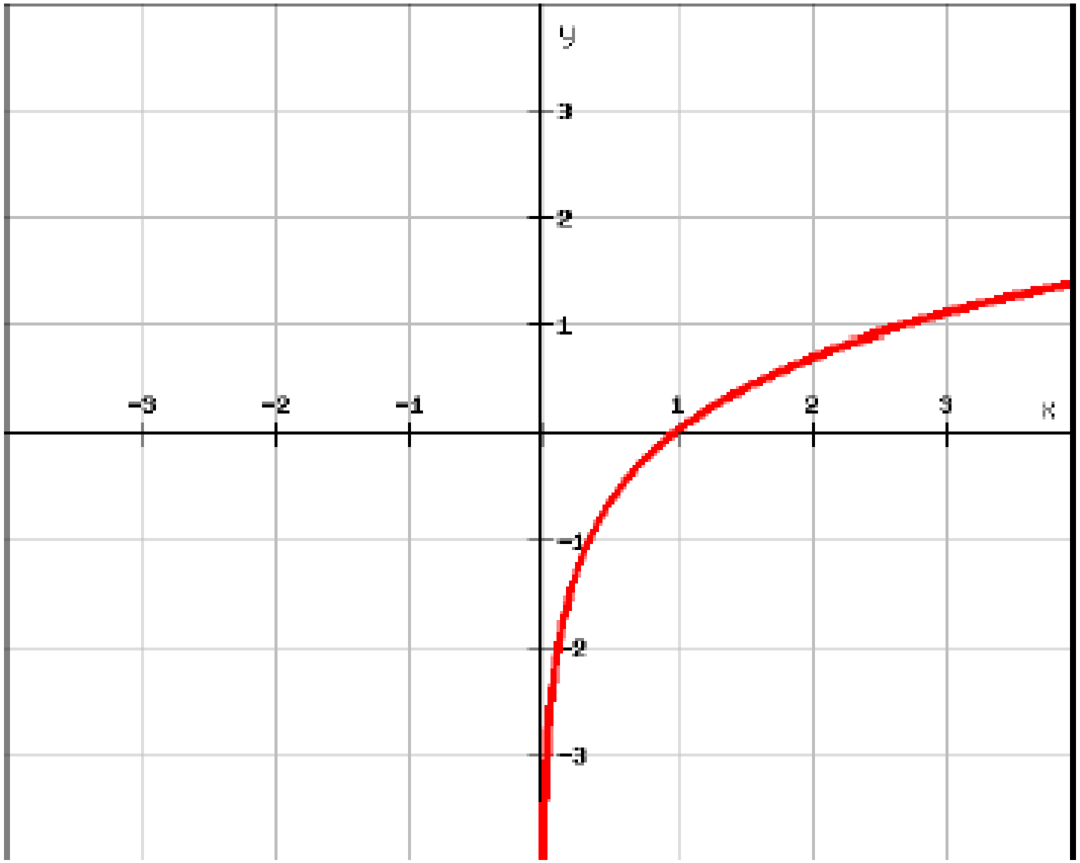
The line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true:

$$\lim_{x \to a} f(x) = \infty \qquad \lim_{x \to a} f(x) = -\infty$$

$$\lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^-} f(x) = -\infty$$

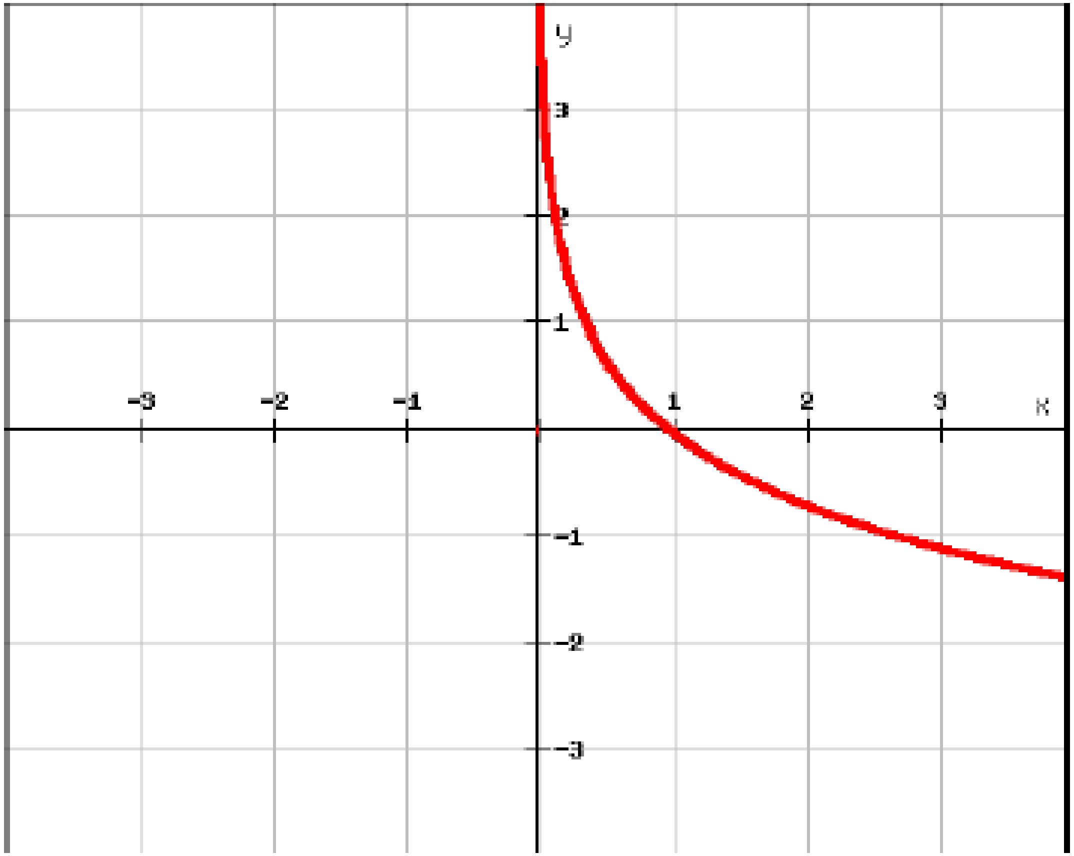
$$\lim_{x \to a^+} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = -\infty$$

$$\lim_{x \to a^+} f(x) = -\infty$$



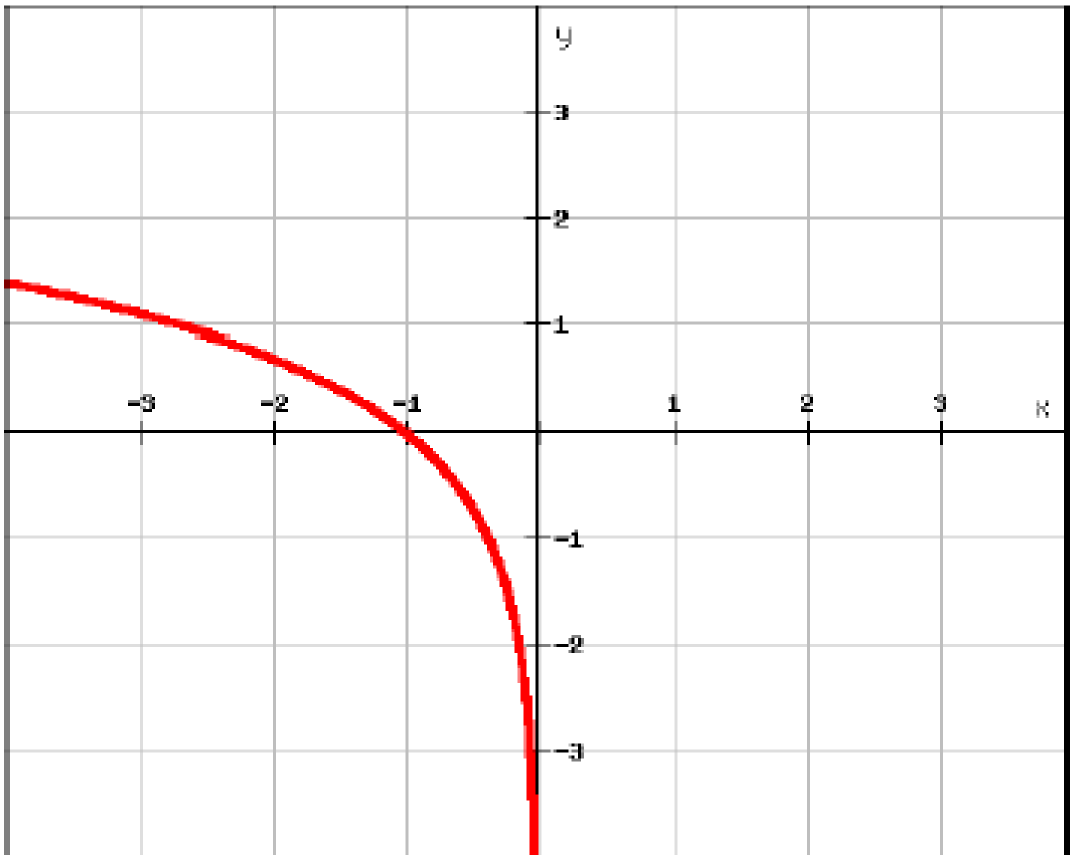
$$f(x) = \ell n(x) \qquad \lim_{x \to 0^+} f(x) = -\infty$$

$$\lim_{x\to a^+} f(x) = \infty$$



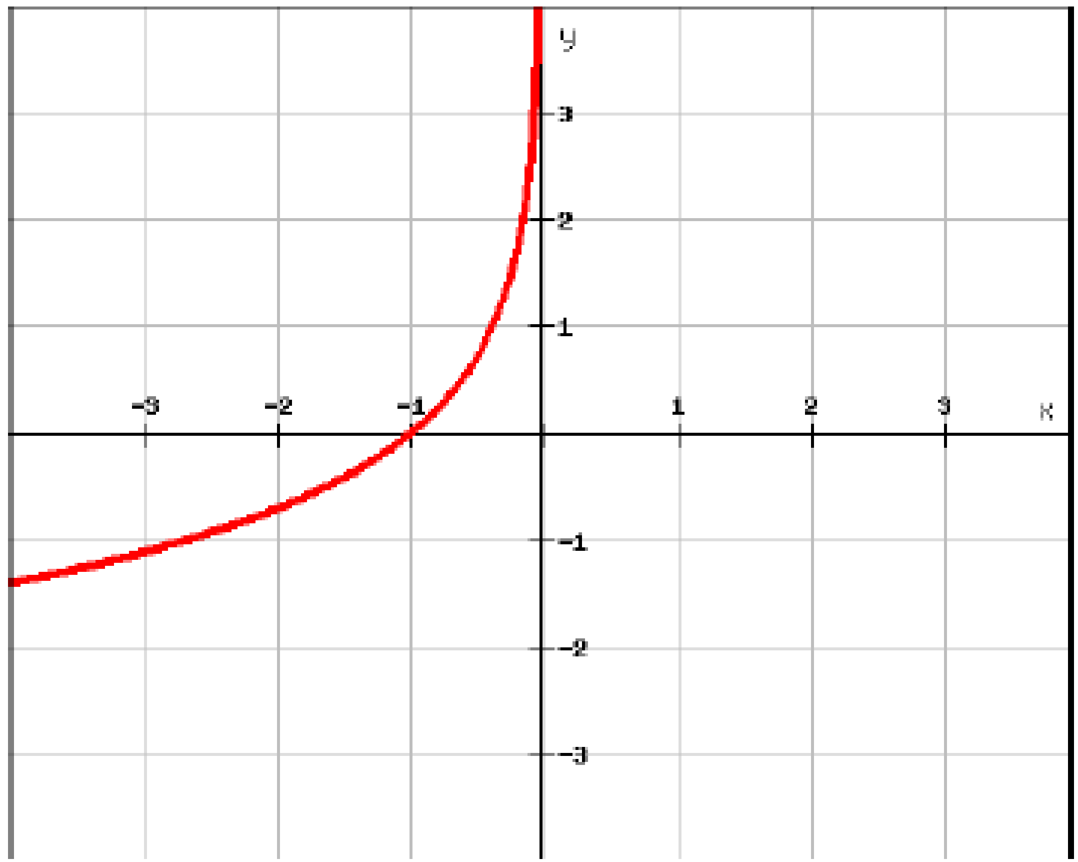
$$f(x) = -\ell n(x) \qquad \lim_{x\to 0^+} -\ell n(x) = \infty$$

$$\lim_{x \to a^-} f(x) = -\infty$$



$$f(x) = \ell n(-x) \qquad \lim_{x \to 0^-} \ell n(-x) = -\infty$$

$$\lim_{x \to a^-} f(x) = \infty$$



$$f(x) = -\ell n(-x) \qquad \lim_{x \to 0^-} -\ell n(-x) = \infty$$

Example

Find the vertical asymptotes of the following function:

$$f(x) = \frac{x^2 + 2}{4x + 4x^2}$$

You should find **two** x values where $f(x)$ might have a vertical asymptote. Call these values a and b. Now find

$$\lim_{x \to a^+} f(x) \qquad \text{and} \qquad \lim_{x \to a^-} f(x)$$

and

$$\lim_{x \to b^+} f(x) \qquad \text{and} \qquad \lim_{x \to b^-} f(x)$$

So far, we have looked very closely at problems like:

$$\lim_{x \to a} f(x)$$

Now we want to consider what happens when

$$x \to \infty \quad \text{and} \quad x \to -\infty$$

That is, we want to see what happens when x gets very large or very small. We can't substitute $x = \infty$ or $x = -\infty$ into our functions, so we must use different techniques.

We read the expression

$$\lim_{x \to \infty} f(x) = L$$

as:

"the limit of $f(x)$ as x tends to infinity is L".

Another way of thinking about

$$\lim_{x \to \infty} f(x) = L$$

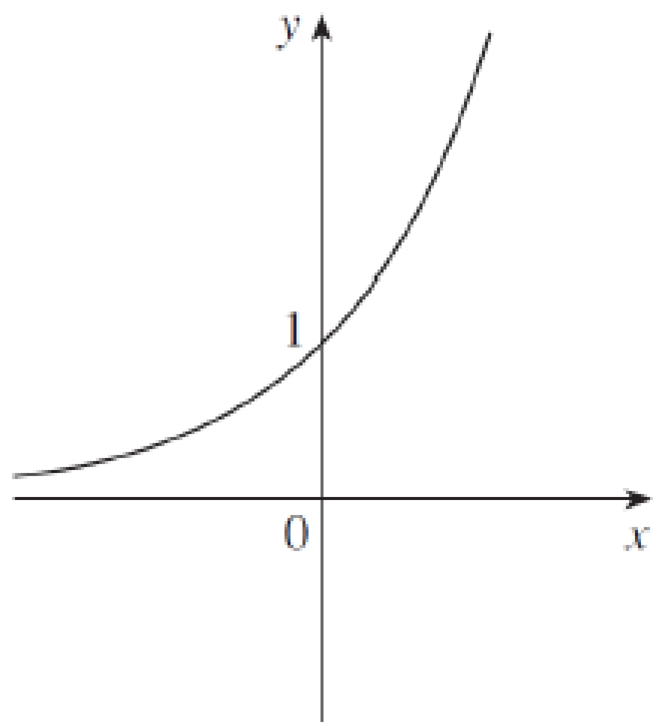
is: the values of $f(x)$ can be made arbitrarily **close** to L by taking x sufficiently **large**.

Very important: when we write

$$\lim_{x \to \infty} f(x) = L$$

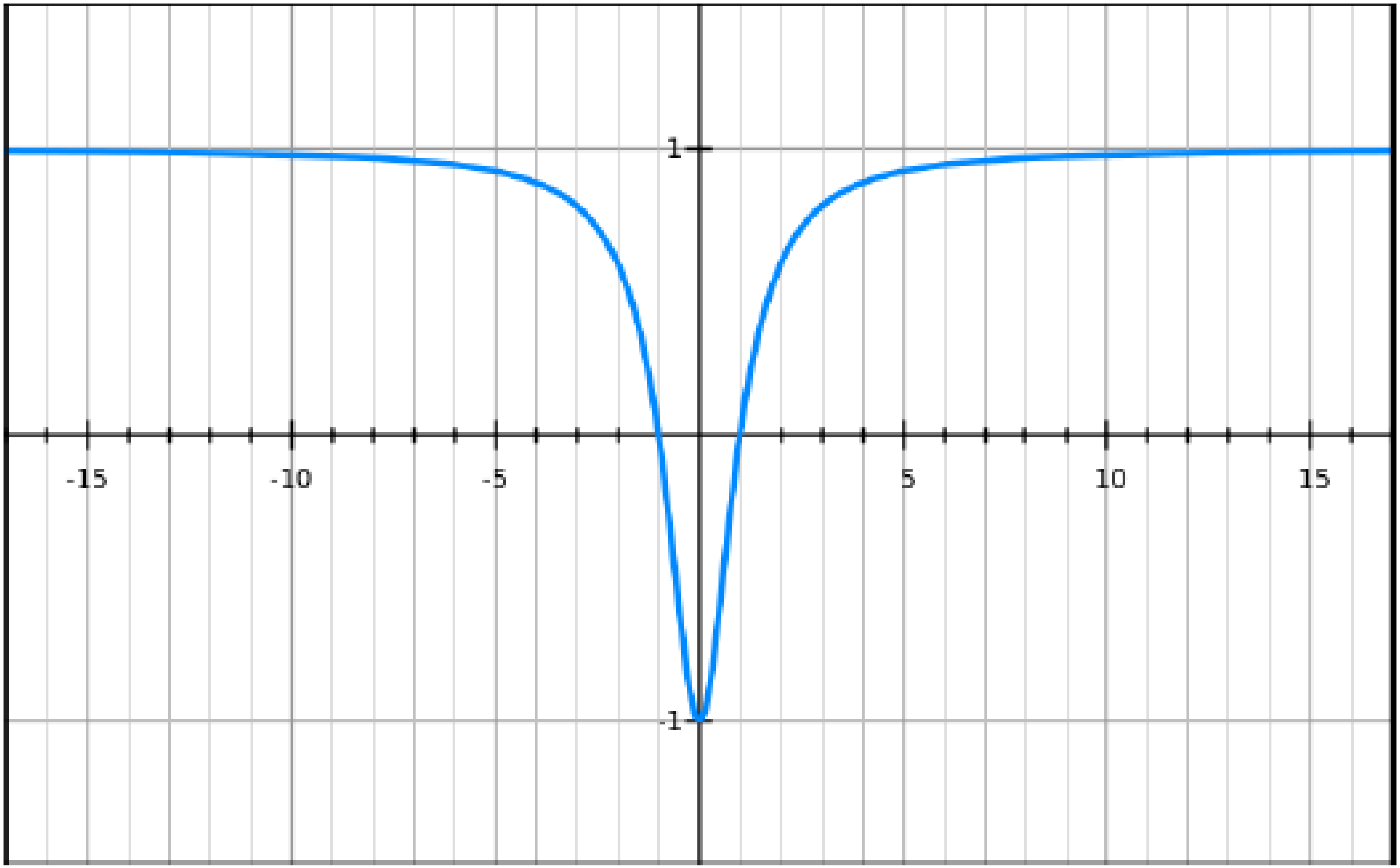
we **are not** saying that $f(\infty) = L$!

Of course we are also interested in the limit of a function as x becomes very small, i.e. as x approaches $-\infty$.

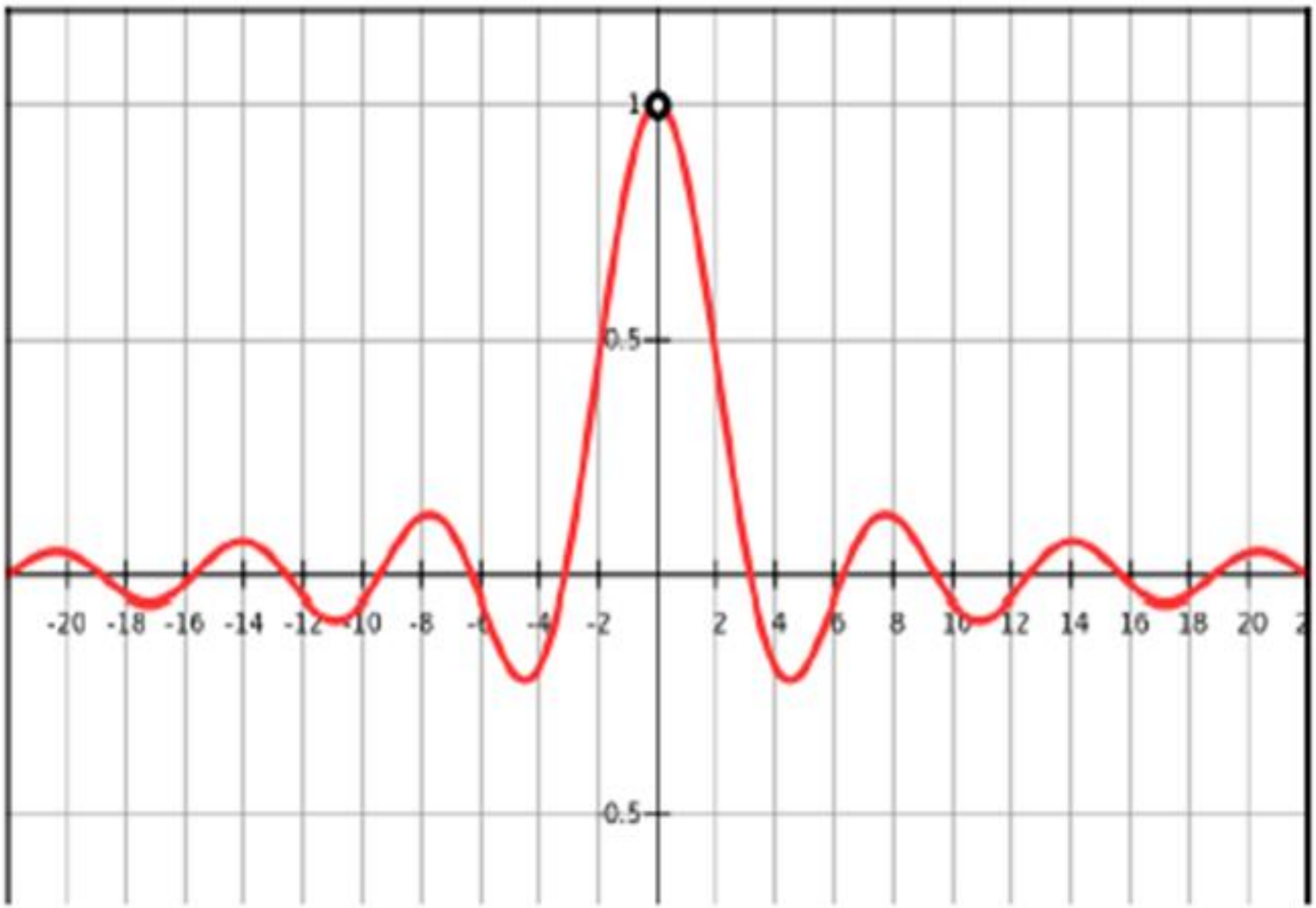


For example: $y = e^x$

$$\lim_{x\to\infty} \frac{x^2-1}{x^2+1} = 1 \quad \text{and} \quad \lim_{x\to-\infty} \frac{x^2-1}{x^2+1} = 1$$



$$\lim_{x\to\infty} \frac{\sin x}{x} = 0 \quad \text{and} \quad \lim_{x\to-\infty} \frac{\sin x}{x} = 0$$



Definition: the line $y = L$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

$$\lim_{x\to\infty} f(x) = L \quad \text{or} \quad \lim_{x\to-\infty} f(x) = L$$

Our main technique for calculating limits as $x \to \infty$ and $x \to -\infty$ is to use the fact that if x is on the bottom of a fraction, then as x increases, the value of the fraction decreases.

Example: $f(x) = \dfrac{1}{x}$

If $x = 10$, then $f(x) = 0.1$
If $x = 100$, then $f(x) = 0.01$
If $x = 1000$, then $f(x) = 0.001$
If $x = 1,000,000$, then $f(x) = 0.000001$

Theorem: If $r > 0$ is a rational number, then

$$\lim_{x \to \infty} \frac{1}{x^r} = 0$$

If $r > 0$ is a rational number such that x^r is defined for all x, then

$$\lim_{x \to -\infty} \frac{1}{x^r} = 0$$

In particular:

$$\lim_{x \to \infty} \frac{1}{x} = 0 \qquad \lim_{x \to \infty} \frac{1}{x^2} = 0 \qquad \lim_{x \to -\infty} \frac{1}{x} = 0$$

Example:

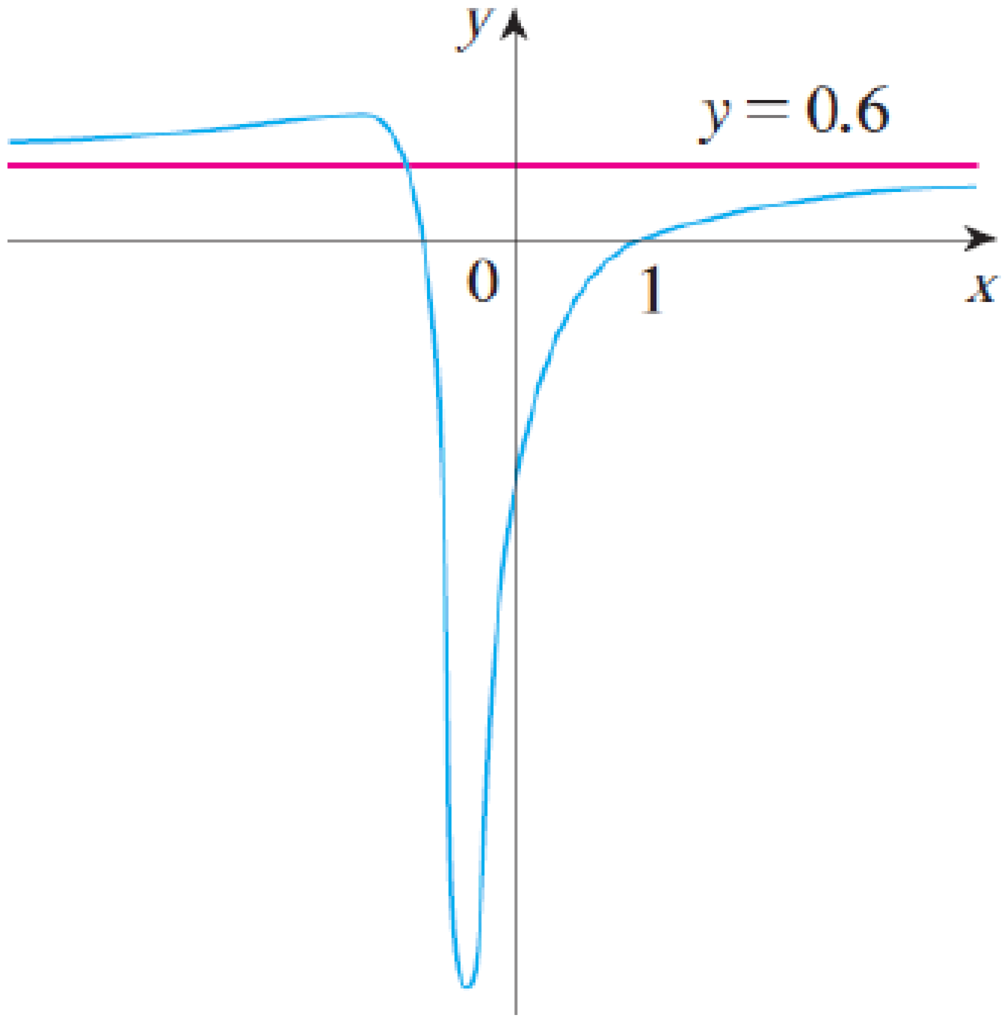
Evaluate

$$\lim_{x\to\infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 1}$$

How do we do this?

To evaluate the limit at infinity of any rational function, we first divide both the numerator and denominator by the highest power of x in the denominator.

$$f(x) = \frac{3x^2 - x - 2}{5x^2 + 4x + 1}$$



Example:

Find the vertical and horizontal asymptotes of the graph of the function

$$f(x) = \frac{\sqrt{2x^2+1}}{3x-5}$$

Note:

$$\text{if } x \geqslant 0 \text{ then } \sqrt{x^2} = x$$

but,

$$\text{if } x < 0 \text{ then } \sqrt{x^2} = -x$$

The graph of $f(x) = \dfrac{1}{x}$

$$f(x) = \frac{\sqrt{2x^2+1}}{3x-5}$$

Note:

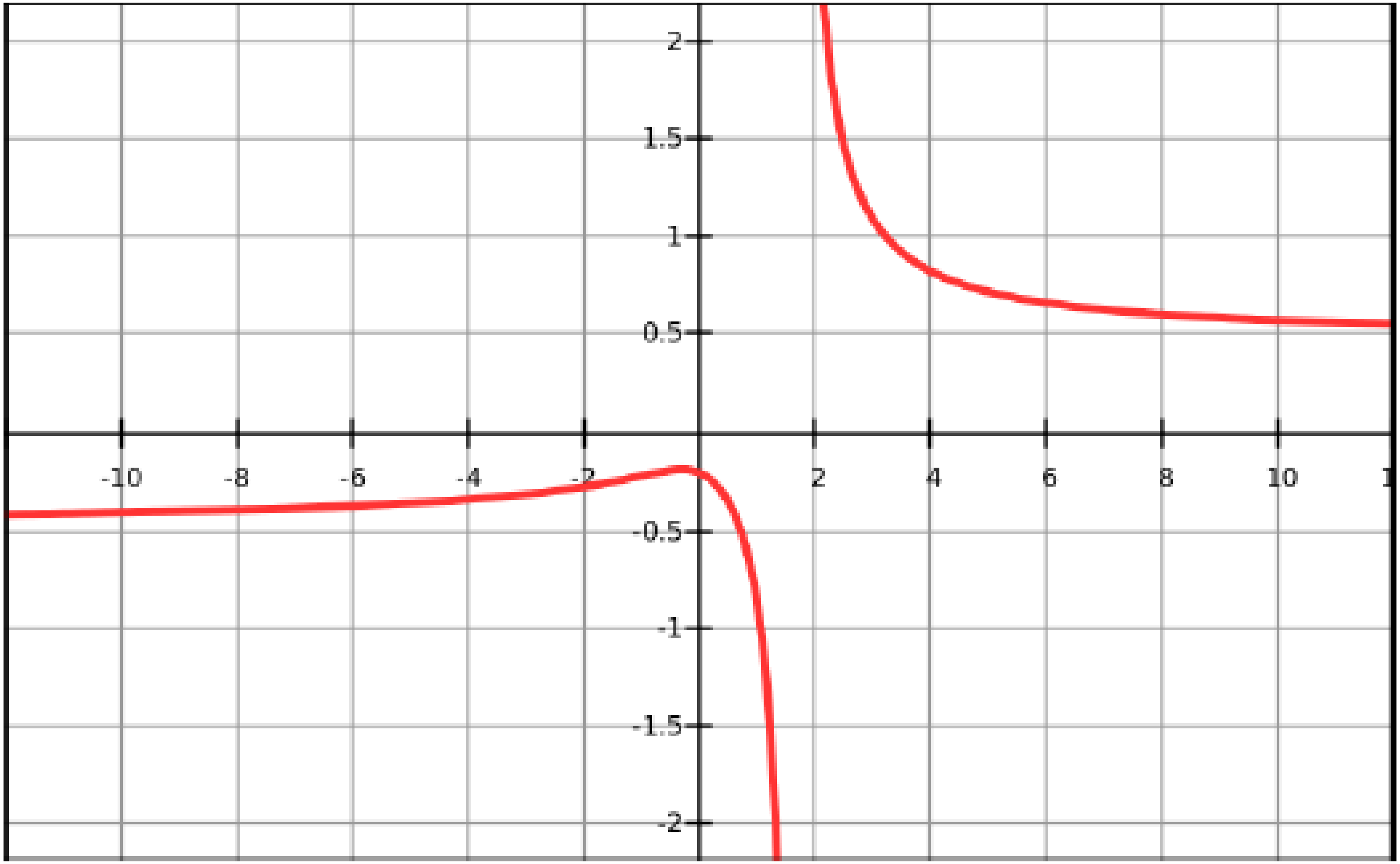
If $x \geqslant 0$ then

$$\frac{\sqrt{g(x)}}{x} = \frac{\sqrt{g(x)}}{\sqrt{x^2}} = \sqrt{\frac{g(x)}{x^2}}$$

If $x < 0$ then

$$\frac{\sqrt{g(x)}}{x} = \frac{\sqrt{g(x)}}{-\sqrt{x^2}} = -\sqrt{\frac{g(x)}{x^2}}$$

The graph of $f(x) = \frac{\sqrt{2x^2+1}}{3x-5}$



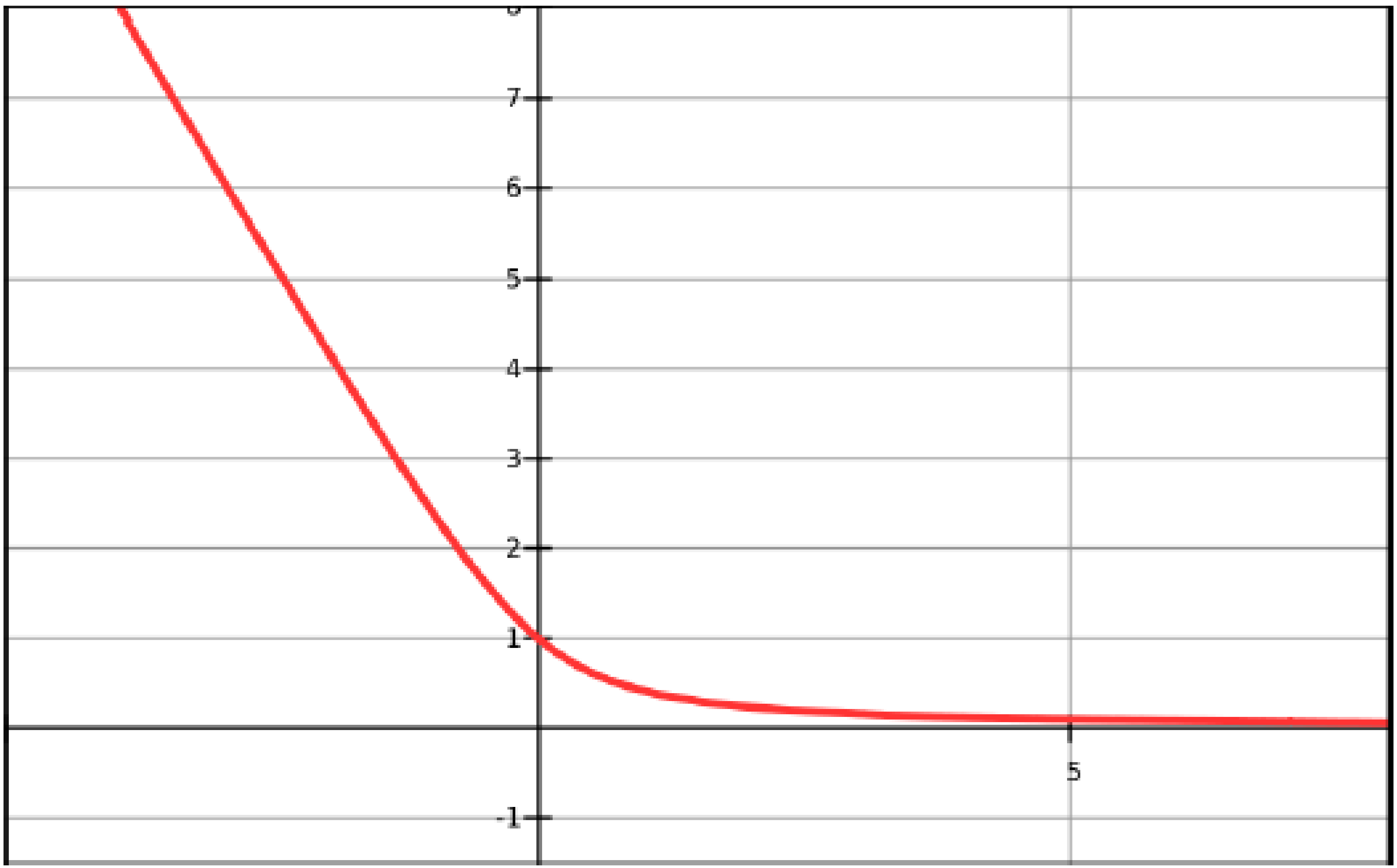
More techniques for calculating limits

What do we do if we have two functions subtracted from one another but cannot tell how they grow relative to one another as $x \to \infty$?

Example: $\lim_{x \to \infty} (\sqrt{x^2 + 1} - x)$

To help us see what is happening as x approaches ∞, we multiply above and below by the radical conjugate.

$$f(x) = \sqrt{x^2 + 1} - x$$



Precise definition: let f be a function defined on some interval (a, ∞). Then

$$\lim_{x \to \infty} f(x) = L$$

means that for every $\varepsilon > 0$ there exists a corresponding N such that:

$$\text{if } x > N \text{ then } |f(x) - L| < \varepsilon.$$

Example: $f(x) = \dfrac{1}{x}$ and $\lim_{x \to \infty} f(x) = 0$

Precise definition: let f be a function defined on some interval (a, ∞). Then

$$\lim_{x \to -\infty} f(x) = L$$

means that for every $\varepsilon > 0$ there exists a corresponding N such that:

$$\text{if } x < N \text{ then } |f(x) - L| < \varepsilon.$$

Infinite limits at infinity

Consider

$$\lim_{x\to\infty} x^3 \qquad \text{and} \qquad \lim_{x\to-\infty} x^3$$

Find $\displaystyle\lim_{x\to\infty}(x^2 - x)$

Find $\displaystyle\lim_{x\to-\infty}(x^3 + 200x^2)$

Find $\displaystyle\lim_{x\to\infty}\frac{x^2 + x}{3 - x}$

Definition - Infinite Limit:

Let f be a function defined on some interval (a, ∞). Then

$$\lim_{x \to \infty} f(x) = \infty$$

means that for every positive number M, there exists N such that:

$$\text{if } x > N \text{ then } f(x) > M.$$

More examples:

- $\lim_{x\to\infty} (\sqrt{x} + \sqrt{x+1})$
- $\lim_{x\to-\infty} \sqrt{2-x}$
- $\lim_{x\to0^+} \left(\frac{1}{\sqrt{x}} - \frac{1}{x}\right)$
- $\lim_{x\to0^+} \left(1 + \frac{1}{x}\right)$

Powers under roots

Consider the following example:

$$\lim_{x\to\infty} \frac{4x^2 - 3x + 6}{\sqrt{x^4 - 7x^3 + 2}}$$

Here the "highest power in the denominator" is x^2 because we have x^4 under a square root.

If we divide above and below by x^2 we can calculate that the limit is 4.

1. Tentukan limit-limit berikut:

a. $\lim_{x\to\infty} \frac{3-2x}{x+5}$

b. $\lim_{x\to\infty} \frac{3x\sqrt{x}+3x+1}{x^2-x+11}$

c. $\lim_{x\to\infty} \frac{2x+1}{\sqrt{x^2+3}}$ $\underline{\Psi}$

d. $\lim_{x\to-\infty} \frac{2x+1}{\sqrt{x^2+3}}$ $\underline{\Psi}$

e. $\lim_{x\to\infty} \left(\sqrt{2x^2+3}-\sqrt{2x^2-5}\right)$

f. $\lim_{x\to-\infty} \frac{9x^3+1}{x^2-2x+2}$

g. $\lim_{x\to 3^+} \frac{3+x}{3-x}$ $\underline{\Psi}$

h. $\lim_{x\to 3^-} \frac{3+x}{3-x}$ $\underline{\Psi}$

i. $\lim_{x\to 0^-} \frac{1+\cos x}{\sin x}$

LIMIT

CONTINUITY

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Definition: A function f is continuous at a number $a \in \mathbb{R}$ if $\lim_{x \to a} f(x) = f(a)$.

Very important: the above definition implies there are **three conditions** that must be satisfied for a function to be continuous at a:

(1) f must be defined at a

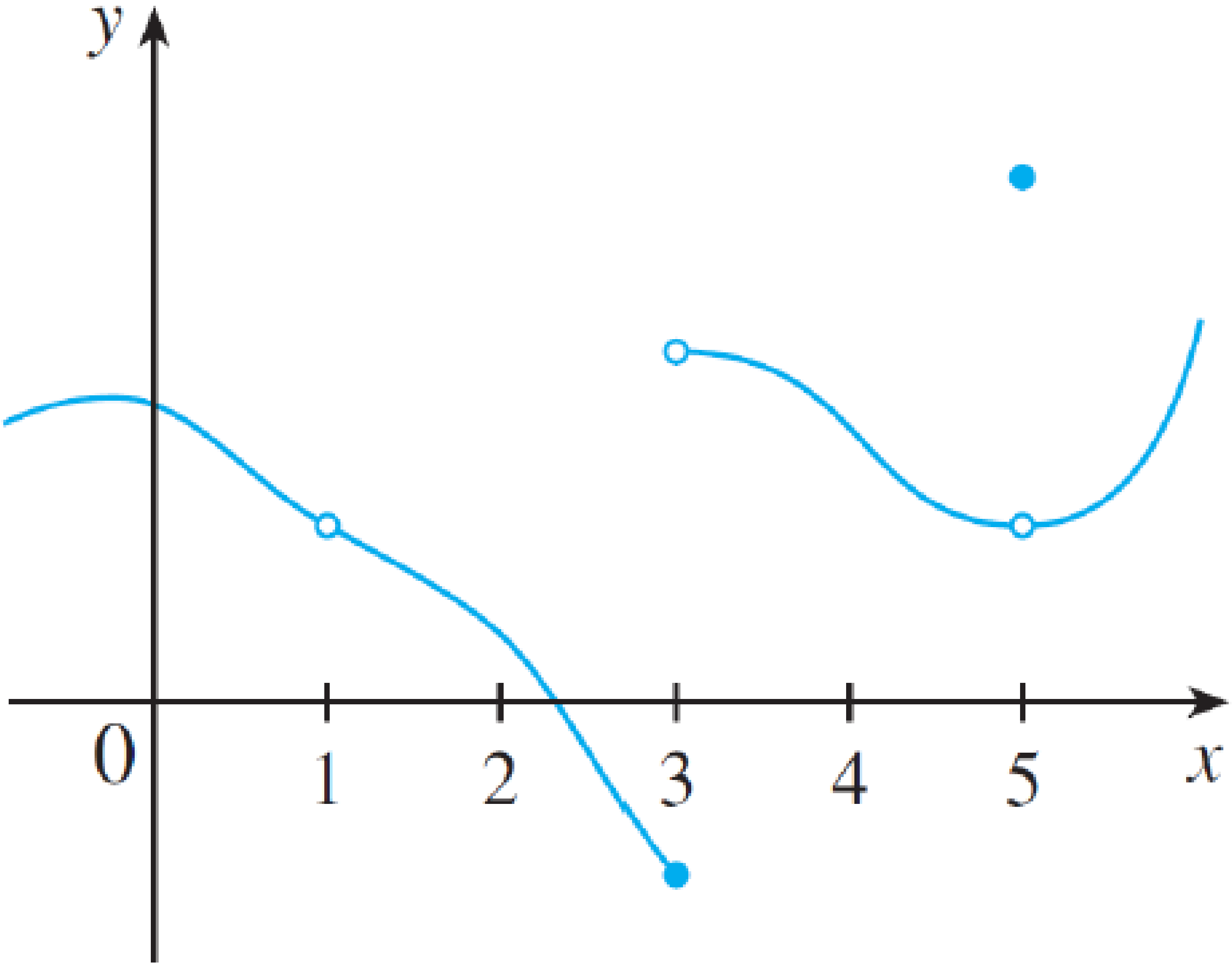
(2) $\lim_{x \to a} f(x)$ must exist

(3) $\lim_{x \to a} f(x) = f(a)$

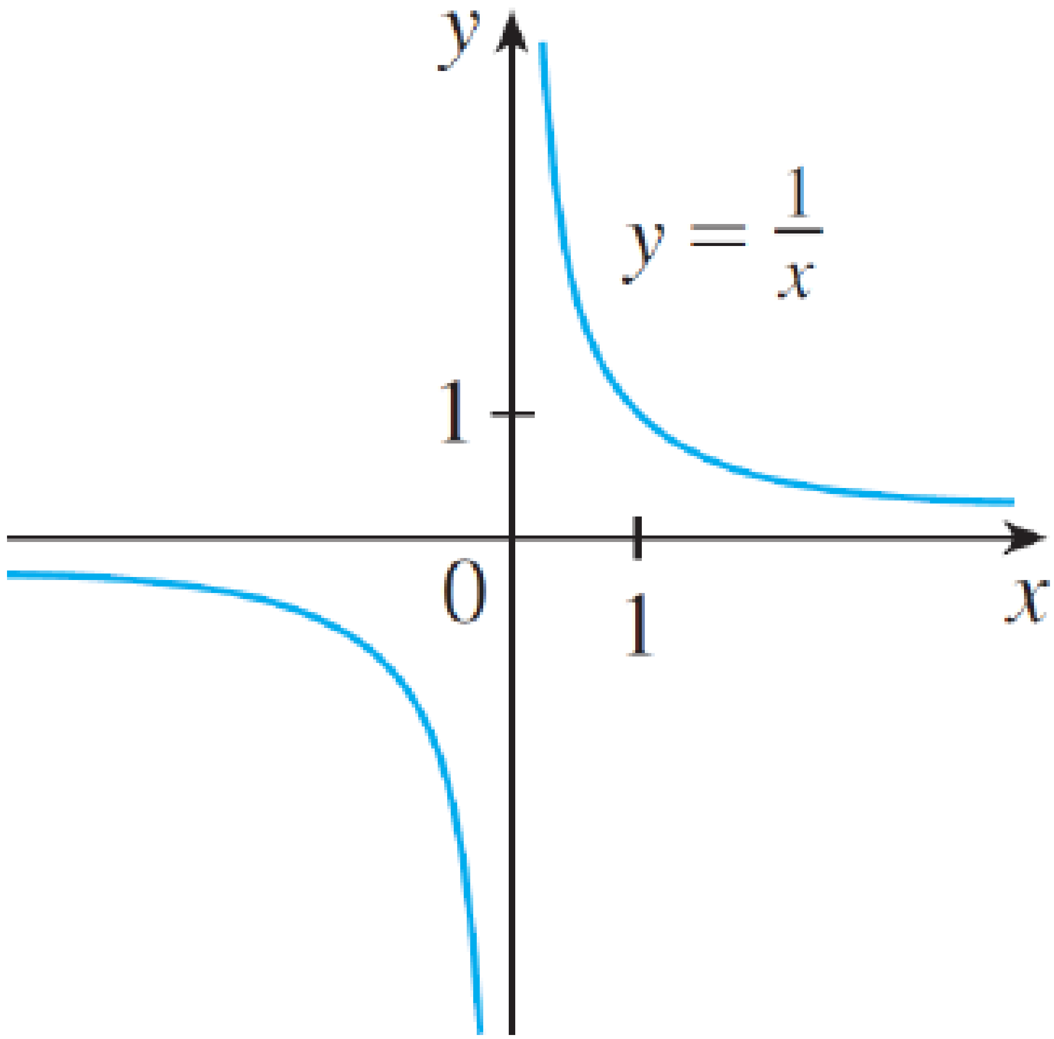
If a function $f(x)$ is not continuous when $x = a$, we say that "f is discontinuous at a".

A point at which a function is discontinuous is called a **discontinuity**.

Examples of discontinuities



More examples of discontinuities



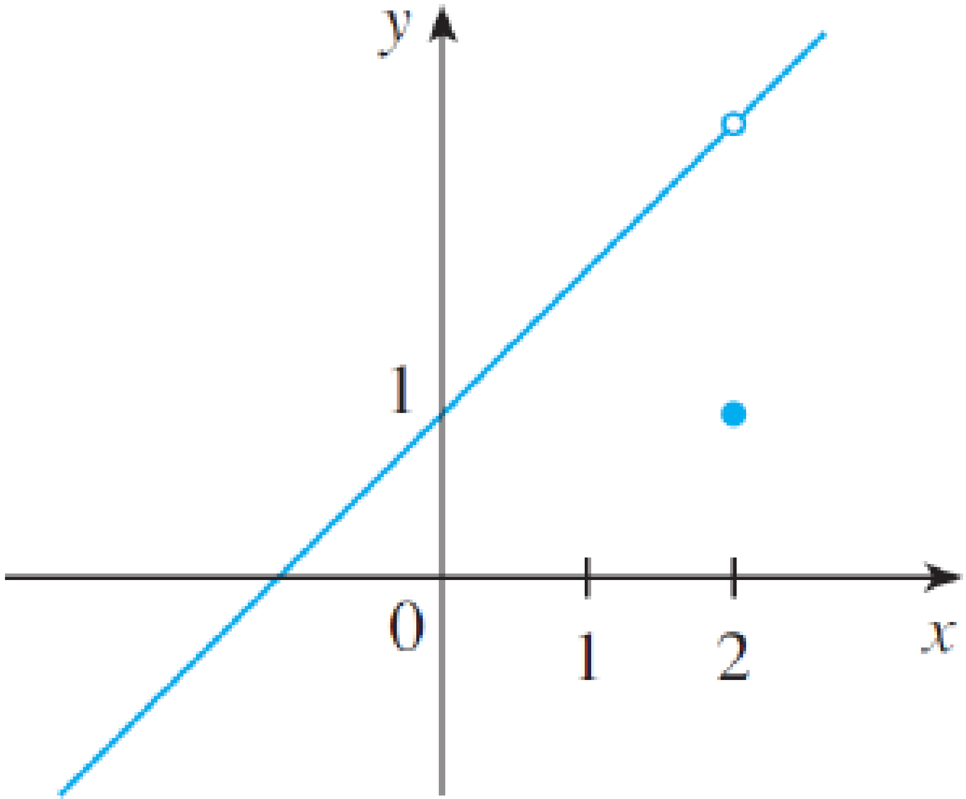
Examples:

Where are the following discontinuous?

(a) $f(x) = \dfrac{x^2 - x - 2}{x - 2}$

(b) $f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$

(c) $f(x) = \begin{cases} \frac{x^2-x-2}{x-2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$



(c) $f(x) = \begin{cases} \dfrac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$

Types of discontinuities

In example **(a)** and example **(c)** we have a **removable discontinuity**. We call them "removable" because if we redefined that point we could "remove" the discontinuity.

The discontinuity in **(b)** is called an **infinite discontinuity**.

One-sided continuity:

Definition: A function f is continuous **from the right** at a if

$$\lim_{x \to a^+} f(x) = f(a)$$

and continuous **from the left** at a if

$$\lim_{x \to a^-} f(x) = f(a)$$

Example of one-sided continuity:

The Heaviside function:

$$H(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geqslant 0 \end{cases}$$

The function is continuous from the right at 0, but not continuous from the left at 0.

This is because

$$\lim_{t \to 0^+} H(t) = 1 = H(0)$$

but

$$\lim_{t \to 0^-} H(t) = 0 \neq H(0)$$

Definition:

A function f is **continuous on an interval** if it is continuous at every number in the interval.

Note: if f is defined only on one side of an endpoint of the interval, we understand "continuous" to mean "continuous from the right" or "continuous from the left".

Example:

Show that $f(x) = 1 - \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$.

To do this we must prove three equalities:

- $\lim_{x \to a} f(x) = f(a)$ for any $a \in (-1, 1)$
- $\lim_{x \to -1^+} f(x) = f(-1)$
- $\lim_{x \to 1^-} f(x) = f(1)$.

Building continuous functions:

Theorem: if f and g are continuous at a and c is a constant, then the following functions are also continuous at a:

1. $f + g$
2. $f - g$
3. cf
4. fg
5. $\dfrac{f}{g}$ if $g(a) \neq 0$

Theorem:

(a) Any polynomial is continuous everywhere; that is, it is continuous on $\mathbb{R} = (-\infty, \infty)$.

(b) Any rational function is continuous wherever it is defined; that is, it is continuous on its domain.

Theorem: the following types of functions are continuous at every number in their domains:

- polynomials
- rational functions
- root functions
- trig functions
- inverse trig functions
- exponential functions
- logarithmic functions

Example:

$$f(x) = \begin{cases} 2x + 8a & \text{if } x < 0 \\ ax^2 - 4b & \text{if } 0 \leqslant x \leqslant 2 \\ -3x + 18 & \text{if } x > 2 \end{cases}$$

What values of a and b make f continuous everywhere?

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